

8

Cardiorespiratory Responses to Acute Exercise

In this chapter and in the web study guide

Cardiovascular Responses to Acute Exercise

Heart Rate

Stroke Volume

Cardiac Output

The Fick Equation

The Cardiac Response to Exercise

Blood Pressure

Blood Flow

Blood

The Integrated Cardiovascular Response to Exercise

AUDIO FOR FIGURE 8.2 describes the use of a submaximal exercise test to estimate maximal exercise capacity.

VIDEO 8.1 presents Ben Levine on physiological differences in trained versus untrained people and the relationship between cardiac output and oxygen use.

AUDIO FOR FIGURE 8.9 describes an example of cardiovascular drift.

ANIMATION FOR FIGURE 8.12 details the integrated cardiovascular response to exercise.

ACTIVITY 8.1 *Cardiovascular Response to Exercise* reviews cardiovascular changes occurring during exercise.

ACTIVITY 8.2 *Cardiovascular Response Scenarios* explores how cardiovascular responses contribute to real-life situations.

Respiratory Responses to Acute Exercise

Pulmonary Ventilation During Dynamic Exercise

Breathing Irregularities During Exercise

Ventilation and Energy Metabolism

Respiratory Limitations to Performance

Respiratory Regulation of Acid–Base Balance

ACTIVITY 8.3 *Pulmonary Ventilation During Exercise* investigates the response of pulmonary ventilation to exercise and the factors that affect the phases of pulmonary ventilation.

ACTIVITY 8.4 *Pulmonary Ventilation and Energy Metabolism* reviews the key terms related to pulmonary ventilation and energy metabolism.

In Closing

Completing a full 26.2 mi (42 km) marathon is a major accomplishment, even for those who are young and extremely fit. On May 5, 2002, Greg Osterman completed the Cincinnati Flying Pig Marathon, his sixth full marathon, finishing in a time of 5 h and 16 min. This is certainly not a world record time, or even an exceptional time for fit runners. However, in 1990 at the age of 35, Greg had contracted a viral infection that went right to his heart and progressed to heart failure. In 1992, he received a heart transplant. In 1993, his body started rejecting his new heart and he also contracted leukemia, not an uncommon response to the antirejection drugs given to transplant patients. He miraculously recovered and started his quest to get physically fit. He ran his first race (15K) in 1994, followed by five marathons in Bermuda, San Diego, New York, and Cincinnati in 1999 and 2001. Greg is an excellent example of both human resolve and physiological adaptability.

After reviewing the basic anatomy and physiology of the cardiovascular and respiratory systems, this chapter looks specifically at how these systems respond to the increased demands placed on the body during acute exercise. With exercise, oxygen demand by the active muscles increases significantly. Metabolic processes speed up and more waste products are created. During prolonged exercise or exercise in a hot environment, body temperature increases. In intense exercise, H^+ concentration increases in the muscles and blood, lowering their pH.

Cardiovascular Responses to Acute Exercise

Numerous interrelated cardiovascular changes occur during dynamic exercise. The primary goal of these adjustments is to increase blood flow to working muscle; however, cardiovascular control of virtually every tissue and organ in the body is also altered. To better understand the changes that occur, we must examine the function of both the heart and the peripheral circulation. In this section, we examine changes in all components of the cardiovascular system from rest to acute exercise, looking specifically at the following:

- Heart rate
- Stroke volume

- Cardiac output
- Blood pressure
- Blood flow
- The blood

We then see how these changes are integrated to maintain adequate blood pressure and provide for the exercising body's needs.

Heart Rate

Heart rate (HR) is one of the simplest physiological responses to measure and yet one of the most informative in terms of cardiovascular stress and strain. Measuring HR involves simply taking the subject's pulse, usually at the radial or carotid artery. Heart rate is a good indicator of relative exercise intensity.

Resting Heart Rate

Resting heart rate (RHR) averages 60 to 80 beats/min in most individuals. In highly conditioned, endurance-trained athletes, resting rates as low as 28 beats/min have been reported. This is mainly due to an increase in parasympathetic (vagal) tone that accompanies endurance exercise training. Resting heart rate can also be affected by environmental factors; for example, it increases with extremes in temperature and altitude.

Just before the start of exercise, preexercise HR usually increases above normal resting values. This is called the anticipatory response. This response is mediated through release of the neurotransmitter norepinephrine from the sympathetic nervous system and the hormone epinephrine from the adrenal medulla. Vagal tone also decreases. Because preexercise HR is elevated, reliable estimates of the true RHR should be made only under conditions of total relaxation, such as early in the morning before the subject rises from a restful night's sleep.

Heart Rate During Exercise

When exercise begins, HR increases directly in proportion to the increase in exercise intensity ([figure 8.1](#)), until near-maximal exercise is achieved. As maximal exercise intensity is approached, HR begins to plateau even as the exercise workload continues to increase. This

indicates that HR is approaching a maximal value. The **maximum heart rate (HR_{max})** is the highest HR value achieved in an all-out effort to the point of volitional fatigue. Once accurately determined, HR_{max} is a highly reliable value that remains constant from day to day. However, this value changes slightly from year to year due to a normal age-related decline.

HR_{max} is often estimated based on age because HR_{max} shows a slight but predictable decrease of about one beat per year beginning at 10 to 15 years of age. Subtracting one's age from 220 beats/min provides a reasonable approximation of one's predicted HR_{max} . However, this is only an estimate—individual values vary considerably from this average value. To illustrate, for a 40-year-old woman, HR_{max} would be estimated to be 180 beats/min ($HR_{max} = 220 - 40$ beats/min). However, 68% of all 40-year-olds have actual HR_{max} values between 168 and 192 beats/min (mean $\pm$ 1 standard deviation), and 95% fall between 156 and 204 beats/min (mean $\pm$ 2 standard deviations). This demonstrates the potential for error in estimating a person's HR_{max} . A similar but more accurate equation has been developed to estimate HR_{max} from age. In this equation, $HR_{max} = 208 - (0.7 \times \text{age})$.¹⁶

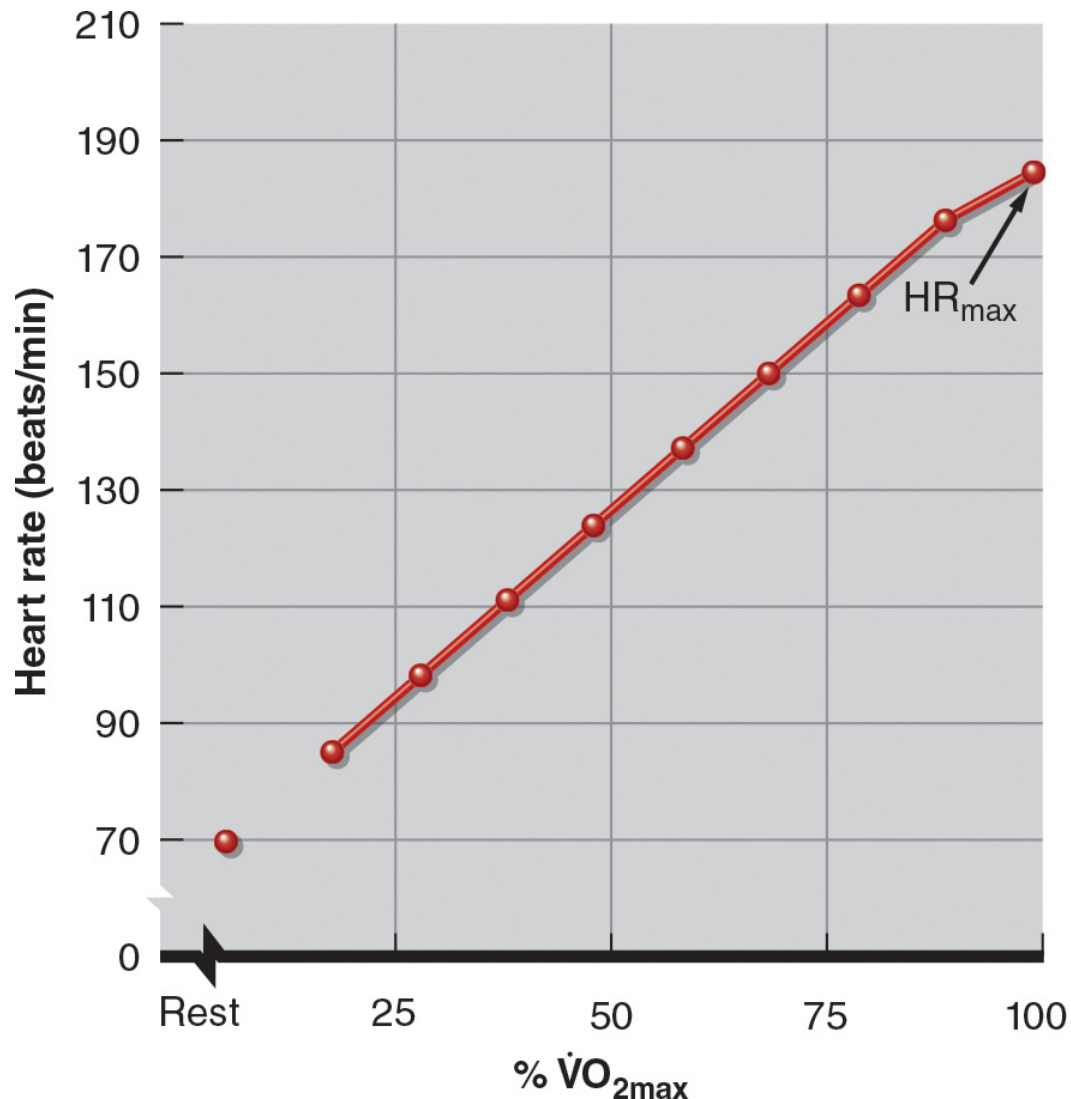


FIGURE 8.1 Changes in heart rate (HR) as a subject progressively walks, jogs, and then runs on a treadmill as intensity is increased. Heart rate is plotted against exercise intensity shown as a percentage of the subject's $\dot{V}O_{2max}$, at which point the rise in HR begins to plateau. The HR at this plateau is the subject's maximal HR or HR_{max}.

When the exercise intensity is held constant at any submaximal workload, HR increases fairly rapidly until it reaches a plateau. This plateau is the **steady-state heart rate**, and it is the optimal HR for meeting the circulatory demands at that specific rate of work. For each subsequent increase in intensity, HR will reach a new steady-state value within 3 min. However, the more intense the exercise, the longer it takes to achieve this steady-state value.

The concept of steady-state heart rate forms the basis for simple exercise tests that have been developed to estimate cardiorespiratory

(aerobic) fitness. In one such test, individuals are placed on an exercise device, such as a cycle ergometer, and then perform exercise at two or three standardized exercise intensities. Those with better cardiorespiratory endurance capacity will have a lower steady-state HR at each exercise intensity than those who are less fit. Thus, a lower steady-state HR at a fixed exercise intensity is a valid predictor of better cardiorespiratory fitness.

Figure 8.2 illustrates results from a submaximal graded exercise test performed on a cycle ergometer by two different individuals of the same age. Steady-state HR is measured at three or four distinct workloads, and a line of best fit is drawn through the data points. Because there is a consistent relation between exercise intensity and energy demand, steady-state HR can be plotted against the corresponding energy ($\dot{V}O_2$) required to do work on the cycle ergometer. The resultant line can be extrapolated to the age-predicted HR_{max} to estimate an individual's maximal exercise capacity. In this figure, subject A has a higher fitness level than subject B because (1) at any given submaximal intensity, this subject's HR is lower and (2) extrapolation to age-predicted HR_{max} yields a higher estimated maximal exercise capacity ($\dot{V}O_{2max}$).

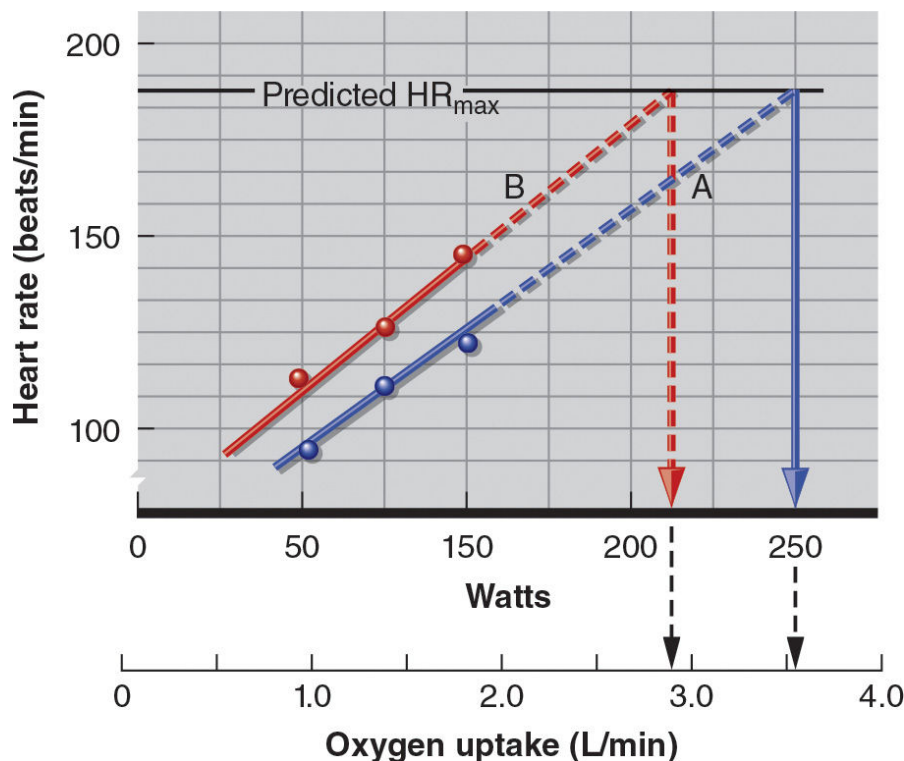
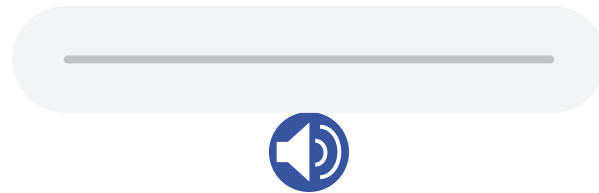


FIGURE 8.2 The increase in heart rate with increasing power output on a cycle ergometer and oxygen uptake is linear within a wide range. The predicted maximal oxygen uptake can be extrapolated using the subject's estimated maximum heart rate as demonstrated here for two subjects with similar estimated maximum heart rates but quite different maximal workloads and $\dot{V}O_{2\max}$ values.

Reprinted by permission from P.O. Åstrand et al., *Textbook of Work Physiology: Physiological Bases of Exercise*, 4th ed. (Champaign, IL: Human Kinetics, 2003), 285.



Heart Rate Variability

Heart rate variability is a measure of the rhythmic fluctuation in HR that occurs because of continuous changes in the sympathetic–parasympathetic balance that controls sinus rhythm. Analysis of HR variability has been used as a method of noninvasively evaluating the relative contributions of the sympathetic and parasympathetic nervous systems at rest and during exercise. During acute aerobic exercise, many different factors contribute to increasing HR variability, including increases in body core temperature, sympathetic nerve activity, and respiratory rate. After a bout of acute exercise, HR variability gradually increases compared to preexercising values due to greater vagal tone. Moreover, changes in HR variability can be used to assess the impact of exercise training (discussed in [chapter 11](#)), the occurrence of overtraining¹⁵ (discussed in [chapter 14](#)), and even as a diagnostic tool in certain clinical populations¹² (discussed in [chapter 20](#)).

RESEARCH PERSPECTIVE 8.1

HUNTING for a Better Prediction of Maximal Heart Rate

Maximal heart rate ($HR_{\max}$) is commonly used in clinical exercise testing and to prescribe exercise intensity in physical training and rehabilitation settings. $HR_{\max}$ can be determined with an individual exercise test to exhaustion and is verified by a plateau of heart rate despite an increase in exercise intensity. However, an exercise test to maximal exertion may not always be feasible, especially in clinical settings where maximal exercise may not be safe or in field tests where advanced equipment (such as a treadmill or stationary

bicycle with adjustable grade or resistance) may not be available. Because of these limitations, there is a need for accurate equations to predict HR_{max} .

HR_{max} declines linearly with age and is estimated using the common formulas in the text of this chapter. However, scientists have suggested that adding other factors, including sex, body mass index (BMI), smoking, and physical activity, to prediction equations may increase their accuracy. In 2013, a group of researchers in Norway set out to develop a new, more accurate prediction formula for HR_{max} .⁸ To do this, the research team studied a subpopulation of participants who were enrolled in the HUNT Fitness Study, a large cohort designed to measure $\dot{V}O_{2max}$ in healthy Norwegian adults. To create a new formula for HR_{max} , the researchers analyzed HR_{max} measured during a peak $\dot{V}O_2$ test, then investigated the relations between HR_{max} and age, sex, physical activity status, BMI, and objectively measured aerobic fitness.

HR_{max} was linearly related to age and was best predicted by the formula

$$HR_{max} = 211 - 0.64 \times \text{age}$$

whereas the traditionally used prediction equation of

$$HR_{max} = 220 - \text{age}$$

(1) overestimated HR_{max} in young individuals, (2) best predicted actual HR_{max} around age 40, and (3) increasingly underestimated HR_{max} as people aged. Unexpectedly, the study team found that HR_{max} was adequately predicted by age alone—accounting for body mass index, sex, smoking status, physical activity, or $\dot{V}O_{2max}$ did not improve the equation's accuracy. This study concluded that the new prediction equation $HR_{max} = 211 - 0.64 \times \text{age}$ most accurately described HR_{max} as a function of age. However, like all prediction formulas, the standard error of ± 11 beats/min must still be taken into consideration. Furthermore, although sex, body mass index, smoking, physical activity, and fitness did not influence the age-related decline in HR_{max} across the sample of subjects they surveyed, these factors may still influence HR_{max} on an individual basis.

This new equation may be better than the quick-and-easy standard $220 - \text{age}$, but direct measurement of HR_{max} using a maximal exercise test is always preferable when possible.

Heart rate, like other signals that repeat periodically over time, can be represented by a power spectrum, which describes how much of the signal occurs at each different frequency. HR signals are analyzed with respect to frequency, rather than time, using a

mathematical technique called *spectral analysis*. In spectral analysis, the variability around the mean HR is separated into the contributing frequency domains. There are many physiological influences on HR variability frequency domains.⁵ Mathematically separating these different elements of HR variability allows researchers to examine the impact of exercise training or disease on each one of the individual contributors. For example, with aerobic exercise training, there is an increase in the parasympathetic control of HR, characterized by greater vagal tone and reduced resting sympathetic nerve activity, that affects the high-frequency domain of HR variability.

Stroke Volume

Stroke volume (SV) also changes during acute exercise to allow the heart to meet the demands of exercise. At near-maximal and maximal exercise intensities, as HR approaches its maximum, SV is a major determinant of cardiorespiratory endurance capacity.

Stroke volume is determined by four factors:

1. The volume of venous blood returned to the heart (the heart can only pump what returns)
2. Ventricular distensibility (the capacity to enlarge the ventricle, to allow maximal filling)
3. Ventricular contractility (the inherent capacity of the ventricle to contract forcefully)
4. Aortic or pulmonary artery pressure (the pressure against which the ventricles must contract)

The first two factors influence the filling capacity of the ventricle, determining how much blood fills the ventricle and the ease with which the ventricle is filled at the available pressure. Together, these factors determine the end-diastolic volume (EDV), sometimes referred to as the **preload**. The last two characteristics influence the ventricle's ability to empty during systole, determining the force with which blood is ejected and the pressure against which it must be expelled into the arteries. The latter factor, the aortic mean pressure, which represents resistance to blood being ejected from the left ventricle (and to a less important extent, the pulmonary artery pressure resistance to flow from the right ventricle), is referred to as

the **afterload**. These four factors combine to determine the SV during acute exercise.

Stroke Volume During Exercise

Stroke volume increases above resting values during exercise. Most researchers agree that SV increases with increasing exercise intensity up to intensities somewhere between 40% and 60% of $\dot{V}O_{2\max}$. At that point, SV typically plateaus, remaining essentially unchanged up to and including the point of exhaustion, as shown in [figure 8.3](#). However, other researchers have reported that SV continues to increase beyond 40% to 60% $\dot{V}O_{2\max}$, even up through maximal exercise intensities, as discussed shortly.

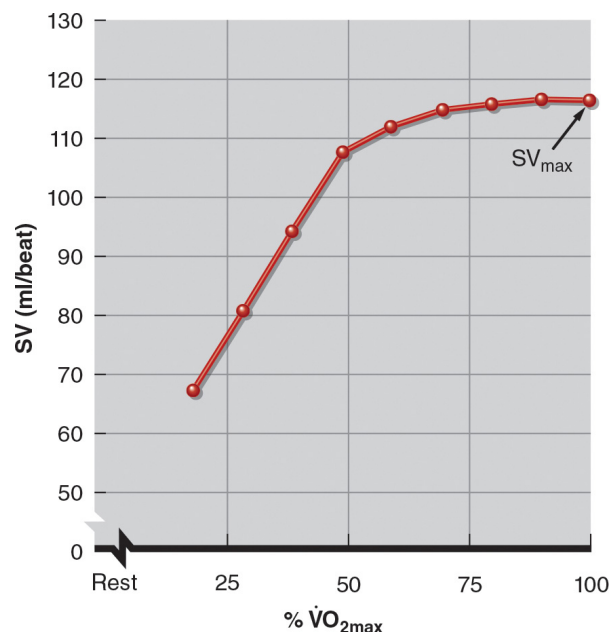


FIGURE 8.3 Changes in stroke volume (SV) as a subject exercises on a treadmill at increasing intensities. Stroke volume is plotted as a function of percent $\dot{V}O_{2\max}$. The SV increases with increasing intensity up to approximately 40% to 60% of $\dot{V}O_{2\max}$, before reaching a maximum ($SV_{\max}$).

When the body is in an upright position, SV can approximately double from resting to maximal values. For example, in active but untrained individuals, SV increases from about 60 to 70 ml/beat at rest to 110 to 130 ml/beat during maximal exercise. In highly trained endurance athletes, SV can increase from 80 to 110 ml/beat at rest to 160 to 200 ml/beat during maximal exercise. During supine exercise,

such as recumbent cycling, SV also increases but usually by only about 20% to 40%—not nearly as much as in an upright position. Why does body position make such a difference?

When the body is in the supine position, blood does not pool in the lower extremities. Blood returns more easily to the heart in a supine posture, which means that resting SV values are higher in the supine position than in the upright position. Thus, the increase in SV with maximal exercise is not as great in the supine position as in the upright position because SV starts out higher. Interestingly, the highest SV attainable in upright exercise is only slightly greater than the resting value in the reclining position. The majority of the SV increase during low to moderate intensities of exercise in the upright position appears to be compensating for the force of gravity that causes blood to pool in the extremities.

Although researchers agree that SV increases as exercise intensity increases up to approximately 40% to 60% $\dot{V}O_{2\max}$, reports about what happens after that point differ. A few studies have shown that SV continues to increase beyond that intensity. Part of this apparent disagreement might result from differences among studies in the mode of exercise testing. Studies that show plateaus in the 40% to 60% $\dot{V}O_{2\max}$ range typically have used cycle ergometers as the mode of exercise. This makes intuitive sense since blood is pooled in the legs during cycle ergometer exercise, resulting in decreased venous return of blood from the legs. Thus, the plateau in SV might be unique to cycling exercise.

Alternatively, in those studies in which SV continued to increase up to maximal exercise intensities, subjects were generally highly trained athletes. Many highly trained athletes, including highly trained cyclists tested on a cycle ergometer, can continue to increase their SV beyond 40% to 60% $\dot{V}O_{2\max}$, perhaps because of adaptations caused by aerobic training. One such adaptation is an increased venous return, which leads to better ventricular filling, and an increased force of contraction (Frank-Starling mechanism). The increases in cardiac output and SV with increasing work as represented by increasing HR, in elite athletes, trained university distance runners, and untrained university students, are illustrated in [figure 8.4](#).

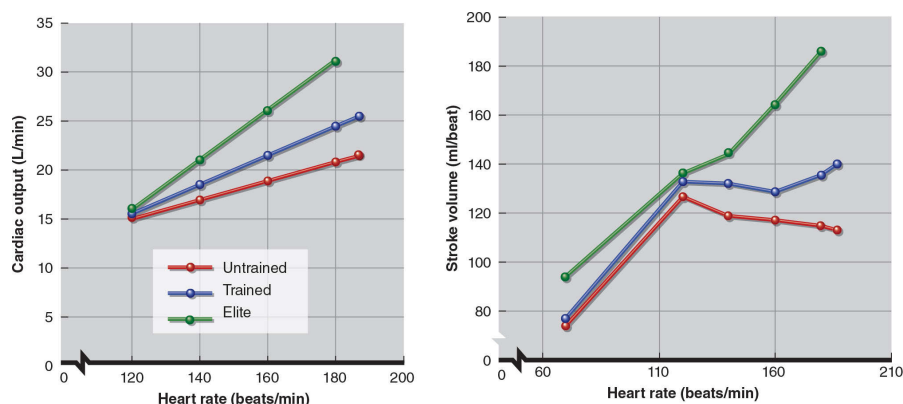


FIGURE 8.4 Cardiac output and stroke volume responses to increasing exercise intensities measured in untrained subjects, trained distance runners, and elite runners.

Adapted by permission from B. Zhou et al., "Stroke Volume Does Not Plateau During Graded Exercise in Elite Male Distance Runners," *Medicine and Science in Sports and Exercise* 33 (2001): 1849-1854.

Importance of Stroke Volume to $\dot{V}O_{2\max}$

$\dot{V}O_{2\max}$ is widely regarded as the single best measure of cardiorespiratory endurance, as discussed in [chapter 5](#). At a maximal exercise intensity, $\dot{V}O_{2\max}$ defines the upper limit of cardiovascular function, that is,

$$\dot{V}O_{2\max} = HR_{\max} \times SV_{\max} \times (a-v)O_{2\max}.$$

[Table 8.1](#) shows the stark difference in $\dot{V}O_{2\max}$ between an elite athlete, a normal age-matched subject, and a cardiac patient with mitral stenosis (a narrowing of the mitral valve). Because differences in $HR_{\max}$ and $(a-v)O_{2\max}$ among these three groups are small, it is the ability to increase SV during maximal exercise that primarily determines $\dot{V}O_{2\max}$.

How Does Stroke Volume Increase During Exercise?

Stroke volume increases during exercise despite the fact that there is less time for ventricular filling, especially at high heart rates. For example, at a resting HR of 70 beats/min, filling time between beats is 0.55 sec. At a HR of 195 beats/min, this interval decreases to 0.12 sec.¹³ How does SV increase in light of less time to fill?

One explanation for the increase in SV with exercise is that the primary factor determining SV is increased preload, or the extent to which the ventricle stretches as it fills with blood, that is, the EDV. When the ventricle stretches more during filling, it subsequently

contracts more forcefully. For example, when a larger volume of blood enters and fills the ventricle during diastole, the ventricular walls stretch to a greater extent. To eject that greater volume of blood, the ventricle responds by contracting more forcefully. This is referred to as the **Frank-Starling mechanism**. At the level of the muscle fiber, the greater the stretch of the myocardial cells, the more actin–myosin cross-bridges are formed, and greater force is developed.

TABLE 8.1 The Importance of Stroke Volume in Determining $\dot{V}O_{2\max}$

Group	$\dot{V}O_{2\max}$ (ml/min)	HR _{max} (beats/min)	SV _{max} (ml/beat)	(a-v)O _{2max} (ml/100 ml)
Athletes	6,250	190	205	16
Normal subjects	3,500	195	112	16
Cardiac patients	1,400	190	43	17

Additionally, SV will increase during exercise if the ventricle's contractility (an inherent property of the ventricle) is enhanced. Contractility can increase by increasing sympathetic nerve stimulation or circulating catecholamines (epinephrine, norepinephrine), or both. An improved force of contraction can increase SV with or without an increased EDV by increasing the ejection fraction. Finally, when mean arterial blood pressure is low, SV is greater since there is less resistance to outflow into the aorta. These mechanisms all combine to determine the SV at any given intensity of dynamic exercise.

Stroke volume is much more difficult to measure than HR. Some clinically used cardiovascular diagnostic techniques have made it possible to determine exactly how SV changes with exercise. Echocardiography (using sound waves) and radionuclide techniques (tagging red blood cells with radioactive tracers) have elucidated how the heart chambers respond to increasing oxygen demands during exercise. With either technique, continuous images of the heart can be taken at rest and up to near-maximal intensities of exercise.

Figure 8.5 illustrates the results of one study of normally active but untrained subjects.⁹ In this study, participants were tested during both supine and upright cycle ergometry at rest and at three exercise intensities, which are depicted on the x-axis of figure 8.5.

When one goes from resting conditions to exercise of increasing intensity, there is an increase in left ventricular EDV (a greater filling or preload), which serves to increase SV through the Frank-Starling

mechanism. There is also a decrease in the left ventricular ESV (greater emptying), indicating an increased force of contraction.

Figure 8.5 shows that both the Frank-Starling mechanism and increased contractility are important in increasing SV during exercise. The Frank-Starling mechanism appears to have its greatest influence at lower exercise intensities, and improved contractile force becomes more important at higher exercise intensities.

Recall that HR also increases with exercise intensity. The plateau or small decrease in left ventricular EDV at high exercise intensities could be caused by a reduced ventricular filling time due to the high HR. One study showed that ventricular filling time decreased from about 500 to 700 ms at rest to about 150 ms at HRs between 150 and 200 beats/min.¹⁷ Therefore, with increasing intensities approaching $\dot{V}O_{2\max}$ (and $HR_{\max}$), the diastolic filling time could be shortened enough to limit filling. As a result, EDV might plateau or even start to decrease.

For the Frank-Starling mechanism to increase SV, left ventricular EDV must increase, necessitating an increased venous return to the heart. As discussed in chapter 6, the muscle pump and respiratory pump both aid in increasing venous return. In addition, redistribution of blood flow and volume from inactive tissues such as the splanchnic and renal circulations increases the available central blood volume.

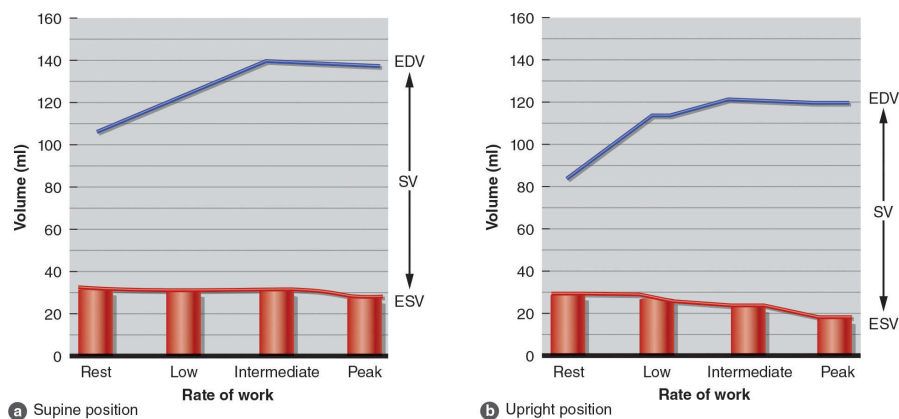


FIGURE 8.5 Changes in left ventricular end-diastolic volume (EDV), end-systolic volume (ESV), and stroke volume (SV) at rest and during low-, intermediate-, and peak-intensity exercise when the subject is in the (a) supine and (b) upright positions. Note that $SV = EDV - ESV$.

Adapted from Poliner et al. (1980).

To review, two factors that can contribute to an increase in SV with increasing intensity of exercise are increased venous return (preload) and increased ventricular contractility. The third factor that contributes to the increase in SV during exercise—a decrease in afterload—results from a decrease in total peripheral resistance. **Total peripheral resistance (TPR)** decreases because of vasodilation of the blood vessels in exercising skeletal muscle. This decrease in afterload allows the left ventricle to expel blood against less resistance, facilitating greater emptying of the ventricle.

Cardiac Output

Since cardiac output is the product of heart rate and stroke volume ($\dot{Q} = \text{HR} \times \text{SV}$), cardiac output predictably increases with increasing exercise intensity (figure 8.6). Resting cardiac output is approximately 5.0 L/min but varies in proportion to the size of the person. Maximal cardiac output varies between less than 20 L/min in sedentary individuals to 40 or more L/min in elite endurance athletes. Maximal $\dot{Q}$ is a function of both body size and endurance training. The linear relationship between cardiac output and exercise intensity is expected because the major purpose of the increase in cardiac output is to meet the muscles' increased demand for oxygen. Like $\dot{V}\text{O}_{2\text{max}}$, when cardiac output approaches maximal exercise intensity, it may reach a plateau (figure 8.6). In fact, it is likely that $\dot{V}\text{O}_{2\text{max}}$ is ultimately limited by the inability of cardiac output to increase further.

VIDEO 8.1 Presents Ben Levine on physiological differences in trained versus untrained people and the relationship between cardiac output and oxygen use.





The Fick Equation

In the 1870s, a cardiovascular physiologist by the name of Adolph Fick developed a principle critical to our understanding of the basic relationship between metabolism and cardiovascular function. In its simplest form, the Fick principle states that the oxygen consumption of a tissue is dependent on blood flow to that tissue and the amount of oxygen extracted from the blood by the tissue. This principle can be applied to the whole body or to regional circulations. Oxygen consumption is the product of blood flow and the difference in concentration of oxygen in the blood between the arterial blood supplying the tissue and the venous blood draining out of the tissue—the $(a-\bar{v})O_2$ difference. Whole-body oxygen consumption ($\dot{V}O_2$) is calculated as the product of the cardiac output ($\dot{Q}$) and $(a-\bar{v})O_2$ difference.

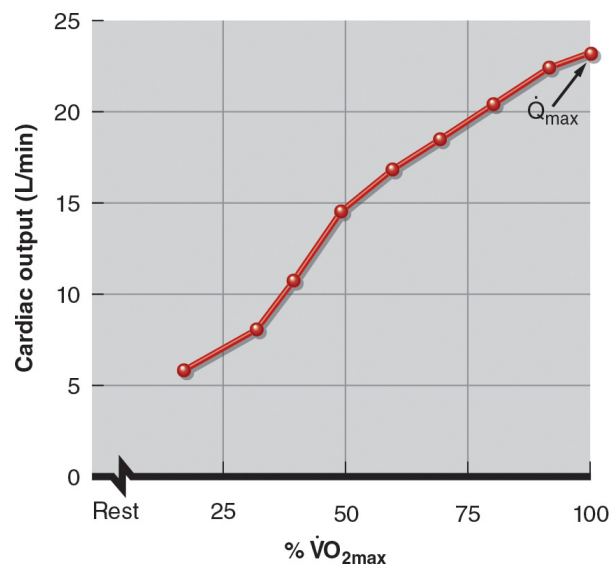


FIGURE 8.6 The cardiac output ($\dot{Q}$) response to walking-running on a treadmill at increasing intensities plotted as a function of percent $\dot{V}O_{2max}$. Cardiac output increases in direct proportion to increasing intensity, eventually reaching a maximum ($\dot{Q}_{max}$).

Fick equation:

$$\dot{V}O_2 = \dot{Q} \times (a-\bar{v})O_2 \text{ difference,}$$

which can be rewritten as

$$\dot{V}O_2 = HR \times SV \times (a-\bar{v})O_2 \text{ difference.}$$

This basic relationship is an important concept in exercise physiology and comes up frequently throughout the remainder of this book.

The Cardiac Response to Exercise

To see how HR, SV, and $\dot{Q}$ vary under various conditions of rest and exercise, consider the following example. An individual first moves from a reclining position to a seated posture and then to standing. Next the person begins walking, then jogging, and finally breaks into a fast-paced run. How does the heart respond?

In a reclining position, HR is ~50 beats/min; it increases to about 55 beats/min during sitting and to about 60 beats/min during standing. When the body shifts from a reclining to a sitting position and then to a standing position, gravity causes blood to pool in the legs, which reduces the volume of blood returning to the heart and thus decreases SV. To compensate for the reduction in SV, HR increases in order to maintain cardiac output; that is, $\dot{Q} = HR \times SV$.

During the transition from rest to walking, HR increases from about 60 to about 90 beats/min. Heart rate increases to 140 beats/min with moderate-paced jogging and can reach 180 beats/min or more with a fast-paced run. The initial increase in HR—up to about 100 beats/min—is mediated by a withdrawal of parasympathetic (vagal) tone. Further increases in HR are mediated by increased activation of the sympathetic nervous system. Stroke volume also increases with exercise, further increasing cardiac output. These relationships are illustrated in [figure 8.7](#).

During the initial stages of exercise in untrained individuals, increased cardiac output is caused by an increase in both HR and SV. When the level of exercise exceeds 40% to 60% of the individual's maximal exercise capacity, SV either plateaus or continues to increase at a much slower rate. Thus, further increases in cardiac output are largely the result of increases in HR. Further SV increases contribute more to the rise in cardiac output at high intensities of exercise in highly trained athletes.

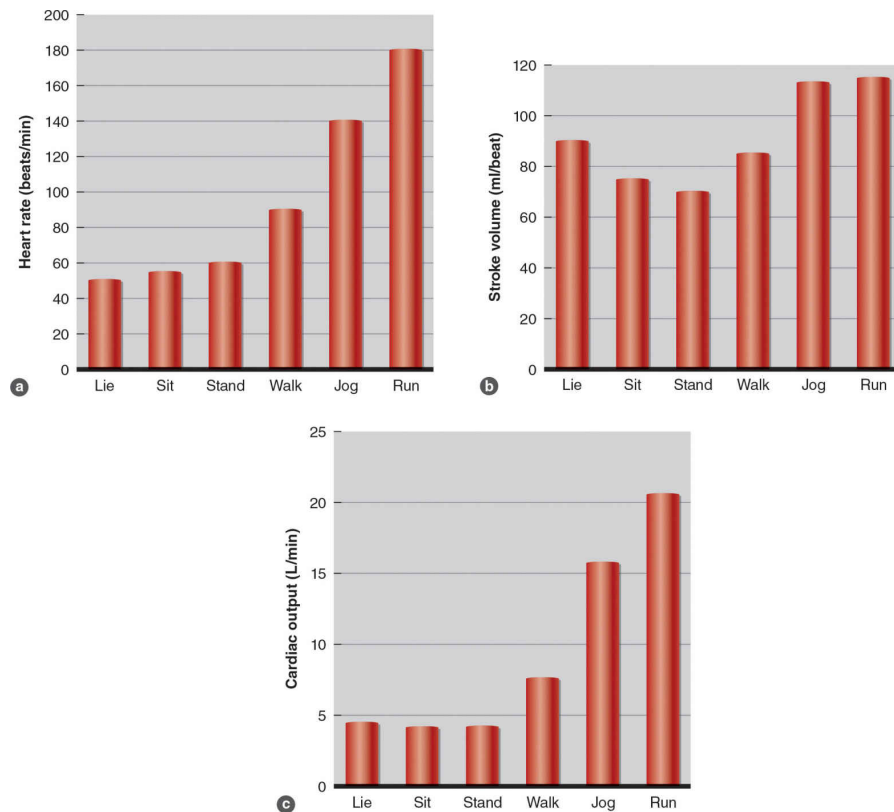


FIGURE 8.7 Changes in (a) heart rate, (b) stroke volume, and (c) cardiac output with changes in posture (lying supine, sitting, and standing upright) and with exercise (walking at 5 km/h [3.1 mph], jogging at 11 km/h [6.8 mph], and running at 16 km/h [9.9 mph]).

Blood Pressure

During endurance exercise, systolic blood pressure increases in direct proportion to the increase in exercise intensity. However, diastolic pressure does not change significantly and may even decrease. As a result of the increased systolic pressure, mean arterial blood pressure increases. A systolic pressure that starts out at 120 mmHg in a normal healthy person at rest can exceed 200 mmHg at maximal exercise. Systolic pressures of 240 to 250 mmHg have been reported in normal, healthy, highly trained athletes at maximal intensities of aerobic exercise.

Increased systolic blood pressure results from the increased cardiac output that accompanies increasing rates of work. This increase in pressure helps facilitate the increase in blood flow through the vasculature. Also, blood pressure (that is, hydrostatic pressure) in large part determines how much plasma leaves the capillaries,

entering the tissues and carrying needed supplies. Thus increased systolic pressure aids substrate delivery to working muscles.

After increasing initially, mean arterial pressure reaches a steady state during submaximal steady-state endurance exercise. As work intensity increases, so does systolic blood pressure. If steady-state exercise is prolonged, the systolic pressure might start to decrease gradually, but diastolic pressure remains constant. The slight decrease in systolic blood pressure, if it occurs, is a normal response and simply reflects increased vasodilation in the active muscles, which decreases the total peripheral resistance (since mean arterial pressure = cardiac output $\times$ total peripheral resistance).

Diastolic blood pressure changes little during submaximal dynamic exercise; however, at maximal exercise intensities, diastolic blood pressure may increase slightly. Remember that diastolic pressure reflects the pressure in the arteries when the heart is at rest (diastole). With dynamic exercise there is an overall increase in sympathetic tone to the vasculature, causing overall vasoconstriction. However, this vasoconstriction is blunted in the exercising muscles by the release of local vasodilators, a phenomenon called functional sympatholysis (discussed in [chapter 6](#)). Thus, because there is a balance between vasoconstriction to inactive regional circulations and vasodilation in active skeletal muscle, diastolic pressure does not change substantially. However, in some cases of cardiovascular disease, increases in diastolic pressure of 15 mmHg or more occur in response to exercise and are one of several indications for immediately stopping a diagnostic exercise test.

Upper body exercise causes a greater blood pressure response than leg exercise at the same absolute rate of energy expenditure. This is most likely attributable to the smaller exercising muscle mass of the upper body compared with the lower body, plus an increased energy demand to stabilize the upper body during arm exercise. This difference in the systolic blood pressure response to upper and lower body exercise has important implications for the heart. Myocardial oxygen uptake and myocardial blood flow are directly related to the product of HR and systolic blood pressure (SBP). This value is referred to as the **rate-pressure product (RPP)**, or double product ($RPP = HR \times SBP$). With static or dynamic resistance exercise or

upper body dynamic exercise, the RPP is elevated, indicating increased myocardial oxygen demand. The use of RPP as an indirect index of myocardial oxygen demand is important in clinical exercise testing.

Periodic blood pressure increases during resistance exercise, such as weightlifting, can be extreme. With high-intensity resistance training, blood pressure can briefly reach 480/350 mmHg. Very high pressures like these are more commonly seen when the exerciser performs a **Valsalva maneuver** to aid heavy lifts. This maneuver occurs when a person tries to exhale while the mouth, nose, and glottis are closed. This action causes an enormous increase in intrathoracic pressure. Much of the subsequent blood pressure increase results from the body's effort to overcome the high internal pressures created during the Valsalva maneuver.

In Review

- Preexercise HR is not a reliable estimate of RHR because of the anticipatory HR response.
- As exercise intensity increases, HR increases proportionately, approaching $HR_{\max}$ near the maximal exercise intensity.
- To estimate $HR_{\max}$:

$$HR_{\max} = 220 - \text{age in years, or}$$

$$HR_{\max} = 208 - (0.7 \times \text{age in years})$$

- Stroke volume (the amount of blood ejected with each contraction) also increases proportionately with increasing exercise intensity but usually achieves its maximal value at about 40% to 60% of $\dot{V}O_{2\max}$ in untrained individuals. Highly trained individuals can continue to increase SV, sometimes up to maximal exercise intensity.
- Increases in HR and SV combine to increase cardiac output. Thus, more blood is pumped during exercise, ensuring that an adequate supply of oxygen and metabolic substrates reaches the exercising muscles and that the waste products of muscle metabolism are cleared away.
- During exercise, cardiac output increases in proportion to exercise intensity to match the need for increased blood flow to exercising muscles.
- According to the Fick equation, whole-body oxygen consumption ($\dot{V}O_2$) is calculated as the product of the cardiac output ($\dot{Q}$) and (a- $\dot{V}$) O_2 difference.

- The ability to increase cardiac output, predominantly driven by increases in stroke volume, is the primary determinant of $\dot{V}O_{2\max}$.
-

Blood Flow

Acute increases in cardiac output and blood pressure during exercise allow for increased total blood flow to the body. These responses facilitate increased blood to areas where it is needed, primarily the exercising muscles. Additionally, sympathetic control of the cardiovascular system redistributes blood so that areas with the greatest metabolic need receive more blood than areas with low demands.

Redistribution of Blood During Exercise

Blood flow patterns change markedly in the transition from rest to exercise. Through the vasoconstrictor action of the sympathetic nervous system on local arterioles, blood flow is redirected away from areas where elevated flow is not essential to those areas that are active during exercise (see [figure 6.11](#)). Only 15% to 20% of the resting cardiac output goes to muscle, but during high-intensity exercise, the muscles may receive 80% to 85% of the cardiac output. This shift in blood flow to the muscles is accomplished primarily by reducing blood flow to the kidneys and the so-called splanchnic circulation (which includes the liver, stomach, pancreas, and intestines). [Figure 8.8](#) illustrates a typical distribution of cardiac output throughout the body at rest and during heavy exercise. Because cardiac output increases greatly with increasing intensity of exercise, the values are shown both as the relative percentage of cardiac output and as the absolute cardiac output going to each regional circulation at rest and at three intensities of exercise.

Although several physiological mechanisms are responsible for the redistribution of blood flow during exercise, they work together in an integrated fashion. To illustrate this, consider what happens to blood flow during exercise, focusing on the primary driver of the response, namely the increased blood flow requirement of the exercising skeletal muscles.

As exercise begins, active skeletal muscles rapidly require increased oxygen delivery. This need is partially met through sympathetic stimulation of vessels in those areas to which blood flow

is to be reduced (e.g., the splanchnic and renal circulations). The resulting vasoconstriction in those areas allows for more of the (increased) cardiac output to be distributed to the exercising skeletal muscles. In the skeletal muscles, sympathetic stimulation to the constrictor fibers in the arteriolar walls also increases; however, local dilator substances are released from the exercising muscle and overcome sympathetic vasoconstriction, producing an overall vasodilation in the muscle (functional sympatholysis).

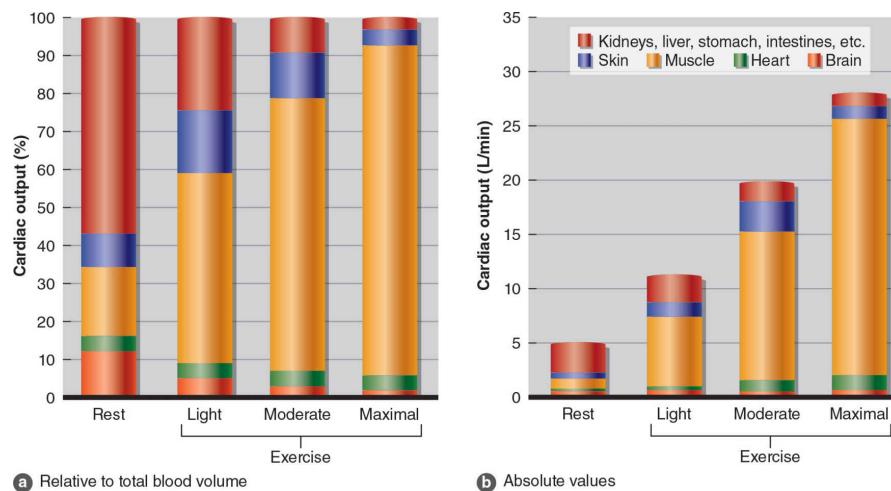


FIGURE 8.8 The distribution of cardiac output at rest and during exercise (a) as a percentage of the total cardiac output and (b) as absolute volumes.

Data from Vander, Sherman, and Luciano (1985).

Many local dilator substances are released in exercising skeletal muscle. As the metabolic rate of the muscle tissue increases during exercise, metabolic waste products begin to accumulate. Increased metabolism causes an increase in acidity (increased hydrogen ions and lower pH), carbon dioxide, and temperature in the muscle tissue. These are some of the local changes that trigger vasodilation of, and increasing blood flow through, the arterioles feeding local capillaries. Local vasodilation is also triggered by the low partial pressure of oxygen in the tissue or a reduction in oxygen bound to hemoglobin (increased oxygen demand), the act of muscle contraction, and possibly other vasoactive substances (including adenosine) released as a result of skeletal muscle contraction.

When exercise is performed in a hot environment, there is also an increase in blood flow to the skin to help dissipate the body heat. The

sympathetic control of skin blood flow is unique in that there are sympathetic vasoconstrictor fibers (similar to skeletal muscle) and sympathetic active vasodilator fibers interacting over most of the skin surface area. During dynamic exercise, as body core temperature rises, there is initially a reduction in sympathetic vasoconstriction, causing a passive vasodilation. Once a specific body core temperature threshold is reached, skin blood flow begins to dramatically increase by activation of the sympathetic active vasodilator system. The increase in skin blood flow during exercise promotes heat loss, because metabolic heat from deep in the body can be released only when blood moves close to the skin. This limits the rate of rise in body temperature, as discussed in more detail in [chapter 12](#).

Cardiovascular Drift

With prolonged aerobic exercise or aerobic exercise in a hot environment at a steady-state intensity, SV gradually decreases and HR increases. Cardiac output is well maintained, but arterial blood pressure also declines. These alterations, illustrated in [figure 8.9](#), have been referred to collectively as **cardiovascular drift**. Cardiovascular drift has traditionally been associated with a progressive increase in the fraction of cardiac output directed to the vasodilated skin to facilitate heat loss and attenuate the increase in body core temperature. With more blood in the skin for the purpose of cooling the body, less blood is available to return to the heart, thus decreasing preload. There is also a small decrease in blood volume resulting from sweating and from a generalized shift of plasma across the capillary membrane into the surrounding tissues. These factors combine to decrease ventricular filling pressure, which decreases venous return to the heart and reduces the EDV. With the reduction in EDV, SV is reduced ($SV = EDV - ESV$). In order to maintain cardiac output ($\dot{Q} = HR \times SV$), HR increases to compensate for the decrease in SV.

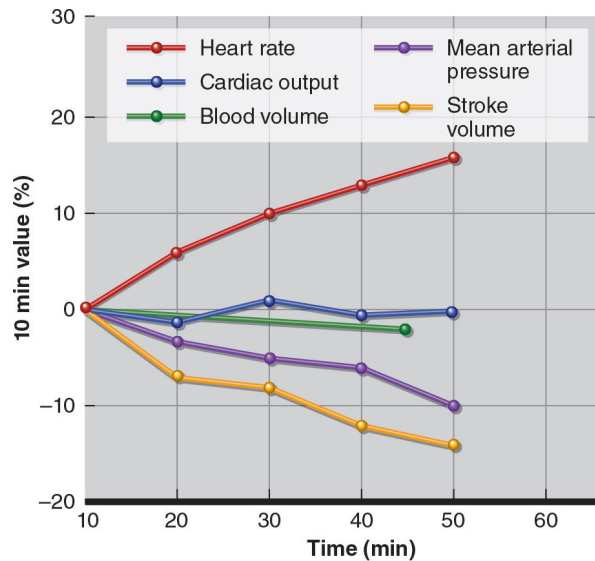
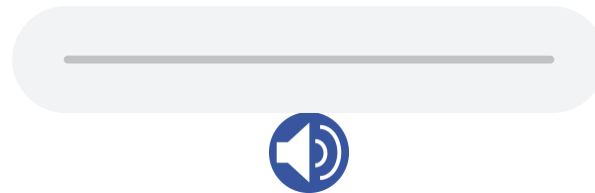


FIGURE 8.9 Circulatory responses to prolonged, moderately intense exercise in the upright posture in a thermoneutral 20 °C environment, illustrating cardiovascular drift. Values are expressed as the percentage of change from the values measured at the 10 min point of the exercise.

Adapted by permission from L.B. Rowell, *Human Circulation: Regulation During Physical Stress* (New York: Oxford University Press, 1986), 230.



A more recent hypothesis has been put forth to explain cardiovascular drift. As HR increases, there is less filling time for the ventricles. This exercise tachycardia may lower SV under the conditions of prolonged exercise even without peripheral displacement of blood volume. From the available research, it is not possible to pinpoint a single hypothesis that fully explains cardiovascular drift, and it is likely that the two mechanisms may interact.

Competition for Blood Supply

When the demands of exercise are added to blood flow demands for all other systems of the body, competition for a limited available cardiac output can occur. This competition for available blood flow can develop among several vascular beds, depending on the specific conditions. For example, there may be competition for available blood

flow between active skeletal muscle and the gastrointestinal system following a meal. McKirnan and coworkers⁷ studied the effects of feeding versus fasting on the distribution of blood flow during exercise in miniature pigs. The pigs were divided into two groups. One group fasted for 14 to 17 h before exercise. The other group ate their morning ration in two feedings: Half the ration was fed 90 to 120 min before exercise and the other half 30 to 45 min before exercise. Both groups of pigs then ran at approximately 65% of their $\dot{V}O_{2\max}$.

Blood flow to the hindlimb muscles during exercise was 18% lower and gastrointestinal blood flow was 23% higher in the fed group than in the fasted group. Similar results in humans suggest that the redistribution of gastrointestinal blood flow to the working muscles is attenuated after a meal. As a practical application, these findings suggest that athletes should be cautious in timing their meals before competition to maximize blood flow to the active muscles during exercise.

Another example of the competition for blood flow is seen in exercise in a hot environment. In this scenario, competition for available cardiac output can occur between the skin circulation for thermoregulation and the exercising muscles. This is discussed in more detail in [chapter 12](#).

Blood

We have now examined how the heart and blood vessels respond to exercise. The remaining component of the cardiovascular system is the blood: the fluid that carries oxygen and nutrients to the tissues and clears away waste products of metabolism. As metabolism increases during exercise, several aspects of the blood itself become increasingly critical for optimal performance.

Oxygen Content

At rest, the blood's oxygen content varies from 20 ml of oxygen per 100 ml of arterial blood to 14 ml of oxygen per 100 ml of venous blood returning to the right atrium. The difference between these two values (20 ml – 14 ml = 6 ml) is referred to as the arterial–mixed venous oxygen difference, or $(a-\bar{v})O_2$ difference. This value represents the extent to which oxygen is extracted, or removed, from the blood as it passes through the body.

With increasing exercise intensity, the $(a-\bar{v})O_2$ difference increases progressively and can almost triple from rest to maximal exercise intensities (see [figure 8.10](#)). This increased difference really reflects a decreasing venous oxygen content, because arterial oxygen content changes little from rest up to maximal exertion. With exercise, more oxygen is required by the active muscles; therefore, more oxygen is extracted from the blood. The venous oxygen content decreases, approaching zero in the active muscles. However, mixed venous blood in the right atrium of the heart rarely decreases below 4 ml of oxygen per 100 ml of blood because the blood returning from the active tissues is mixed with blood from inactive tissues as it returns to the heart. Oxygen extraction by the inactive tissues is far lower than in the active muscles.

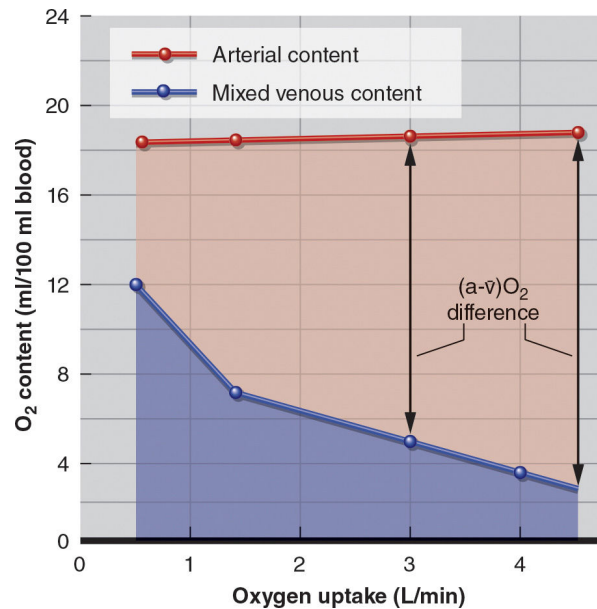
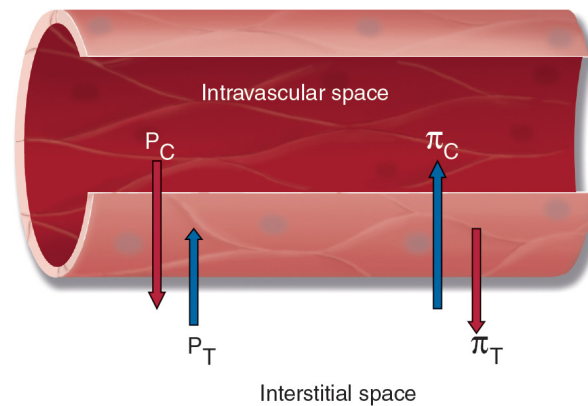


FIGURE 8.10 Changes in the oxygen content of arterial and mixed venous blood and the $(a-\bar{v})O_2$ difference (arterial–mixed venous oxygen difference) as a function of exercise intensity.



$$\text{Net capillary filtration} = (P_C + \pi_T) - (P_T - \pi_C)$$

FIGURE 8.11 Filtration of plasma from the microvasculature. Both the blood pressure (P_C) inside the blood vessel and the oncotic pressure (π_T) in the tissue cause plasma to flow from the intravascular space to the interstitial space. The pressure that the tissue (P_T) exerts on the blood vessel and the oncotic pressure of the blood (π_C) inside the blood vessel cause plasma to be reabsorbed. Net filtration of plasma can be determined by summing the outward forces ($P_C + \pi_T$) and subtracting the inward forces ($P_T - \pi_C$); net capillary filtration = $(P_C + \pi_T) - (P_T - \pi_C)$.

Plasma Volume

Upon standing, or with the onset of exercise, there is an almost immediate loss of plasma from the blood to the interstitial fluid space. The movement of fluid out of the capillaries is dictated by the pressures inside the capillaries, which include the **hydrostatic pressure** exerted by increased blood pressure and the **oncotic pressure**, the pressure exerted by the proteins in the blood, mostly albumin. The pressures that influence fluid movement outside the capillaries are the pressure provided by the surrounding tissue as well as the oncotic pressures from proteins in the interstitial fluid (figure 8.11). Osmotic pressures, those exerted by electrolytes in solution on both sides of the capillary wall, also play a role. As blood pressure increases with exercise, the hydrostatic pressure within the capillaries increases. This increase in blood pressure forces water from the intravascular compartment to the interstitial compartment. Also, as metabolic waste products build up in the active muscle, intramuscular osmotic pressure increases, which draws fluid out of the capillaries to the muscle.

Approximately a 10% to 15% reduction in plasma volume can occur with prolonged exercise, with the largest falls occurring during

the first few minutes. During resistance training, the plasma volume loss is proportional to the intensity of the effort, with similar transient losses of fluid from the vascular space of 10% to 15%.

If exercise intensity or environmental conditions cause sweating, additional plasma volume losses may occur. Although the major source of fluid for sweat formation is the interstitial fluid, this fluid space will be diminished as sweating continues. This increases the oncotic (since proteins do not move with the fluid) and osmotic (since sweat has fewer electrolytes than interstitial fluid) pressures in the interstitial space, causing even more plasma to move out of the vascular compartment into the interstitial space. Intracellular fluid volume is impossible to measure directly and accurately, but research suggests that fluid is also lost from the intracellular compartment during prolonged exercise and even from the red blood cells, which may shrink in size.



A reduction in plasma volume can impair performance. For long-duration activities in which dehydration occurs and heat loss is a problem, blood flow to active tissues may be reduced to allow increasingly more blood to be diverted to the skin in an attempt to lose body heat. Note that a decrease in muscle blood flow occurs only in conditions of dehydration and only at high intensities. Severely reduced plasma volume also increases blood viscosity, which can impede blood flow and thus limit oxygen transport, especially if the hematocrit exceeds 60%.

In activities that last a few minutes or less, body fluid shifts are of little practical importance. As exercise duration increases, however, body fluid changes and temperature regulation become important for

performance. For the football player, the Tour de France cyclist, or the marathon runner, these processes are crucial, not only for competition but also for survival. Deaths have occurred from dehydration and hyperthermia during, or as a result of, various sport activities. These issues are discussed in detail in [chapter 12](#).

Hemoconcentration

When plasma volume is reduced, hemoconcentration occurs. When the fluid portion of the blood is reduced, the cellular and protein portions represent a larger fraction of the total blood volume; that is, they become more concentrated in the blood. This hemoconcentration increases red blood cell concentration substantially—by up to 25%. Hematocrit can increase from 40% to 50%. However, the total number and volume of red blood cells do not change substantially.

The net effect, even without an increase in the total number of red blood cells, is to increase the number of red blood cells per unit of blood; that is, the cells are more concentrated. As the red blood cell concentration increases, so does the blood's per-unit hemoglobin content. This substantially increases the blood's oxygen-carrying capacity, which is advantageous during exercise and provides a distinct advantage at altitude, as discussed in [chapter 13](#).

The Integrated Cardiovascular Response to Exercise

As is evident from all of the changes in cardiovascular function that take place during exercise, the cardiovascular system is extremely complex but responds exquisitely to deliver oxygen to meet the demands of exercising muscle. [Figure 8.12](#) is a simplified flow diagram that illustrates how the body integrates all these cardiovascular responses to provide for its needs during exercise. Key areas and responses are labeled and summarized to help illustrate how these complex control mechanisms are coordinated. It is important to note that although the body attempts to meet the blood flow needs of the muscle, it can do so only if blood pressure is not compromised. Maintenance of arterial blood pressure appears to be the highest priority of the cardiovascular system, regardless of exercise, the environment, or other competing needs.

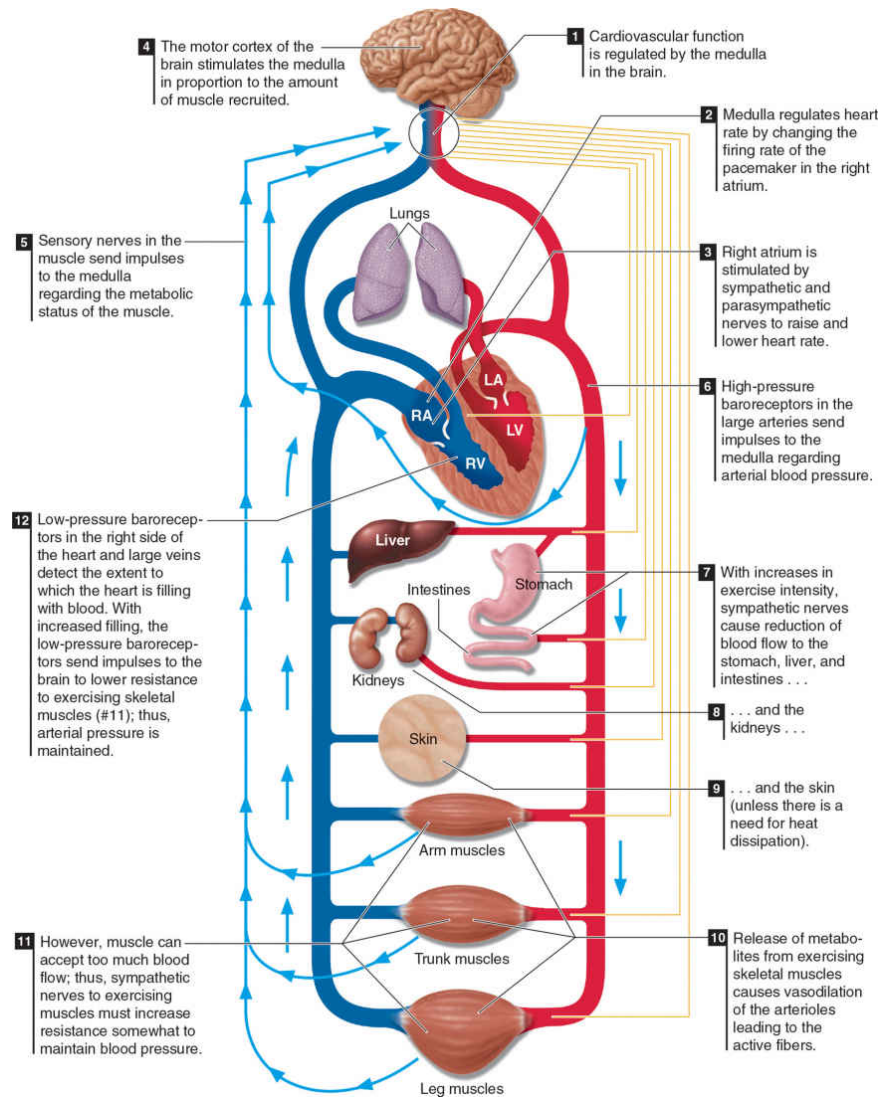


FIGURE 8.12 The integrated cardiovascular response to exercise.

Adapted by permission from E.F. Coyle, "Cardiovascular Function During Exercise: Neural Control Factors," *Sports Science Exchange* 4, no. 34 (1991): 1-6. Adapted with permission of Stokely-Van Camp, Inc.

RESEARCH PERSPECTIVE 8.2

Is Recovery a Distinct Cardiovascular State?

Exercise recovery refers to the time period immediately following a bout of exercise. This period continues until the system has completely recovered, or returned to a resting state, and can last anywhere from seconds to hours depending on the mode and intensity of the exercise. Exercise recovery can also refer to the specific physiological state that exists after exercise, which is distinctly different from the physiology of exercise or the physiology at rest. Some of these physiological changes during recovery may be necessary for long-term adaptation to exercise training, yet some can lead to cardiovascular instability during recovery. Over the last 20 years, the scientific understanding of exercise recovery as a distinct physiological state has grown immensely, mainly through human studies of cardiovascular variables such as blood pressure, heart rate, and cardiac output immediately following aerobic exercise or resistance exercise.¹¹

In general, there is a dose-dependent effect of exercise intensity and duration on the cardiovascular changes that follow aerobic exercise. In general, the increase in vascular conductance (or decrease in resistance due to vasodilation of the blood vessels in the muscle) is greater than the increase in cardiac output following a bout of aerobic exercise. This means that peripheral vasodilation is the driving force that lowers blood pressure after exercise. This reduced blood pressure after exercise is called *postexercise hypotension*, and can last for several hours following a bout of aerobic exercise. The sustained postexercise vasodilation occurs largely within the previously active skeletal muscle, with a smaller but still relevant vasodilation in the nonactive skeletal muscle beds. Blood flow to the other tissues (e.g., brain, gut) reverts more quickly to resting values. Vasodilation of the nonactive skeletal muscle probably occurs due to a resetting of the blood pressure set point at the brain, while vasodilation of the previously active skeletal muscle is due to the release of local vasodilatory molecules. Recently, it has been demonstrated that the one important molecule released by the previously active muscle is histamine. Histamine is elevated in the muscle following exercise, and postexercise vasodilation is reduced by 80% when the actions of histamine are inhibited. While the lasting effects of histamine improve our understanding of what causes postexercise hypotension during recovery, the exercise-related trigger for histamine release from the muscle remains unknown.

The cardiovascular changes that occur during recovery following a bout of resistance exercise are distinctly different from those following acute aerobic exercise. Like aerobic exercise, blood pressure is reduced following resistance exercise. However, in contrast to aerobic exercise, postexercise hypotension following resistance exercise is due to decreases in cardiac output, not to vasodilation in the vascular beds of the previously active muscle. It is unclear why the mechanisms controlling blood pressure during recovery are different

between aerobic and resistance exercise, but these differences are likely due to both central regulation (how the sympathetic nervous system controls blood pressure) and local cellular changes in the muscle. Interestingly, knee-extension exercise that replicates resistance training does not generate a local increase in histamines, while knee-extension exercise that replicates aerobic exercise does. Because combined aerobic and resistance exercise programs do not further reduce postexercise blood pressure compared to aerobic exercise alone, there is probably some overlap in the central mechanisms. Overall, there are fewer studies of the control of postexercise hypotension following resistance exercise. Although the recent research points to a larger role for changes in the central control of blood pressure during recovery, this is an area that requires further study.

Exercise recovery can be viewed as both a window of opportunity for the positive adaptations to training to be manipulated and a vulnerable period in which individuals are at heightened risk for adverse events such as fainting. Fully understanding this period may provide insight into when the cardiovascular system has recovered from prior training and is physiologically ready for additional training stress. The future may include training methods that take advantage of the exercise recovery state to avoid negative consequences of overtraining and to optimize training and health outcomes.

The cardiovascular and respiratory adjustments to dynamic exercise are profound and rapid. Within 1 s of the initiation of muscle contraction, HR dramatically increases by vagal withdrawal and respiration increases. Increases in cardiac output and blood pressure increase blood flow to the active skeletal muscle to meet its metabolic demands. What causes these extremely rapid early changes in the cardiovascular system, since they take place well before metabolic needs of working muscle occur?

Over the years there has been considerable debate over what causes the cardiovascular system to be turned on at the onset of exercise. One explanation is the theory of **central command**, which involves parallel coactivation of both the motor and the cardiovascular control centers of the brain. Activation of central command rapidly increases HR and blood pressure. In addition to central command, the cardiovascular responses to exercise are modified by mechanoreceptors, chemoreceptors, and baroreceptors. As discussed in [chapter 6](#), baroreceptors are sensitive to stretch and send information back to the cardiovascular control centers about blood pressure. Signals from the periphery are sent back to the

cardiovascular control centers through the stimulation of mechanoreceptors that are sensitive to the stretch of the skeletal muscle and through the chemoreceptors that are sensitive to an increase in metabolites in the muscle. Feedback about blood pressure and the local muscle environment helps to fine-tune and adjust the cardiovascular response. These relationships are illustrated in [figure 8.13](#).

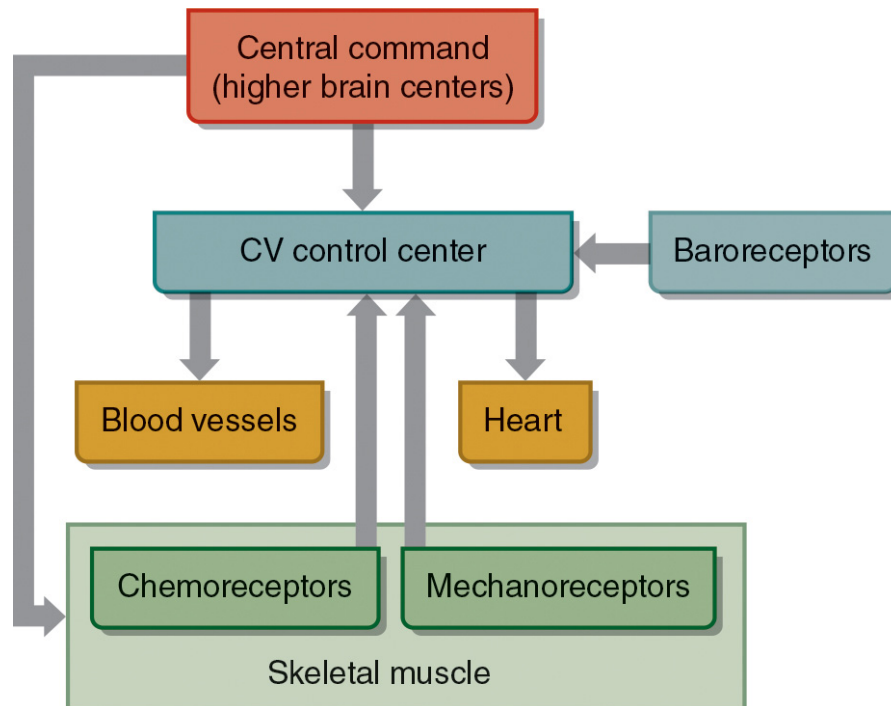


FIGURE 8.13 A summary of cardiovascular (CV) control during exercise.

Adapted by permission from S.K. Powers & E.T. Howley, *Exercise Physiology: Theory and Application to Fitness and Performance*, 5th ed. (New York, McGraw-Hill, 2004), 188. © The McGraw-Hill Education.

In Review

- Mean arterial blood pressure increases immediately in response to exercise, and the magnitude of the increase is proportional to the intensity of exercise. During whole-body endurance exercise, this is accomplished primarily by an increase in systolic blood pressure, with minimal changes in diastolic pressure.
- Systolic blood pressure can exceed 200 to 250 mmHg at maximal exercise intensity, the result of increases in cardiac output. Upper body exercise causes a greater blood pressure response than leg exercise at the same absolute rate of

energy expenditure, likely due to the smaller muscle mass involved and the need to stabilize the trunk during dynamic arm exercise.

- Blood flow is redistributed during exercise from inactive or low-activity tissues of the body like the liver and kidneys to meet the increased metabolic needs of exercising muscles.
 - With prolonged aerobic exercise, or aerobic exercise in the heat, SV gradually decreases and HR increases proportionately to maintain cardiac output. This is referred to as cardiovascular drift and is associated with a progressive increase in blood flow to the vasodilated skin and losses of fluid from the vascular space.
 - The changes that occur in the blood during exercise include the following:
 1. The $(a-\bar{v})O_2$ difference increases as venous oxygen concentration decreases, reflecting increased extraction of oxygen from the blood for use by the active tissues.
 2. Plasma volume decreases. Plasma is pushed out of the capillaries by increased hydrostatic pressure as blood pressure increases, and fluid is drawn into the muscles by the increased oncotic and osmotic pressures in the muscle tissues, a by-product of metabolism. With prolonged exercise or exercise in hot environments, increasingly more plasma volume is lost through sweating.
 3. Hemoconcentration occurs as plasma volume (water) decreases. Although the actual number of red blood cells stays relatively constant, the relative number of red blood cells per unit of blood increases, which increases oxygen-carrying capacity.
-

Respiratory Responses to Acute Exercise

Now that we have discussed the role of the cardiovascular system in delivering oxygen to the exercising muscle, we examine how the respiratory system responds to acute dynamic exercise.

Pulmonary Ventilation During Dynamic Exercise

The onset of exercise is accompanied by an immediate increase in ventilation. In fact, like the HR response, the marked increase in breathing may occur even before the onset of muscular contractions—that is, it may be an anticipatory response. This is shown in [figure 8.14](#) for light, moderate, and heavy exercise. Because of its rapid onset, this initial respiratory adjustment to the demands of exercise is undoubtedly neural in nature, mediated by respiratory control centers

in the brain (central command), although neural signals also come from receptors in the exercising muscle.

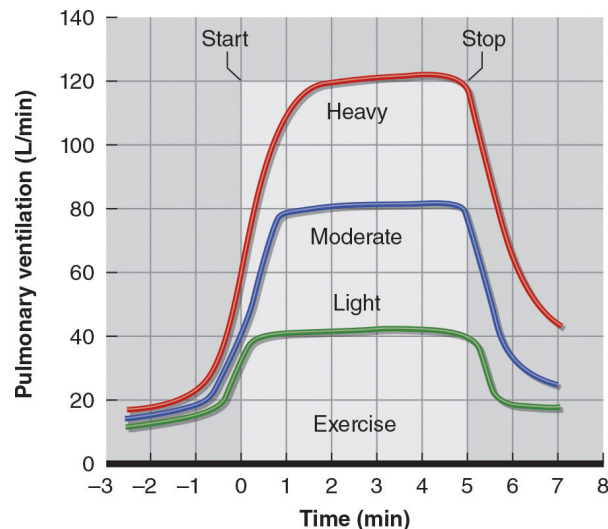


FIGURE 8.14 The ventilatory response to light, moderate, and heavy exercise. The subject exercised at each of the three intensities for 5 min. After an initial steep increase, the ventilation rate tended to plateau at a steady-state value at the light and moderate intensities but continued to increase somewhat at the heavy intensity.

The more gradual second phase of the respiratory increase shown during heavy exercise in [figure 8.14](#) is controlled primarily by changes in the chemical status of the arterial blood. As exercise progresses, increased metabolism in the muscles generates more CO_2 and H^+ . Recall that these changes shift the oxyhemoglobin saturation curve rightward, enhancing oxygen unloading in the muscles, which increases the $(a-\bar{v})\text{O}_2$ difference. Increased CO_2 and H^+ are sensed by chemoreceptors primarily located in the brain, carotid bodies, and lungs, which in turn stimulate the inspiratory center, increasing rate and depth of respiration. Chemoreceptors in the muscles themselves might also be involved. In addition, receptors in the right ventricle of the heart send information to the inspiratory center so that increases in cardiac output can stimulate breathing during the early minutes of exercise. The influences of CO_2 and H^+ concentrations in the blood on breathing rate and pattern serve to fine-tune the neutrally mediated respiratory response to exercise in order to precisely match oxygen delivery with aerobic demands without overtaxing respiratory muscles.

RESEARCH PERSPECTIVE 8.3

Posture Affects Ventilation During Recovery After Exercise

Body posture affects cardiopulmonary function due to the effects of gravity. For example, in the upright posture, the mechanical actions of the inspiratory muscles expand the chest wall and elevate the rib cage against gravity, while changing to the supine posture increases abdominal pressure on the pleural cavity and inspiration is achieved predominantly through abdominal expansion. Despite reports that cardiopulmonary function is affected by body position, few exercise physiologists have investigated the effect of posture during recovery from aerobic exercise.

A 2017 study conducted in Korea examined cardiopulmonary function in relation to body position during recovery from a maximal exercise test.⁴ Subjects were randomly assigned to one of three recovery postures: supine, sitting, or sitting with the trunk leaning forward. Each subject performed a maximal exercise test to exhaustion, then immediately assumed their assigned recovery position. Oxygen uptake, minute ventilatory volume, respiration rate, and heart rate were measured during the assigned posture at rest before the test and at 1, 3, and 5 min of recovery. No differences in these variables were seen preexercise. While there were no differences in heart or respiratory rate between recovery postures, the $\dot{V}O_2$ and minute ventilatory volume were significantly lower during recovery in the group assigned to the trunk-leaning-forward posture. This forward-leaning posture improves ventilatory capacity during recovery from maximal exercise, which, in turn, enables rapid recovery of the respiratory system after exercise. The study team concluded that the forward-leaning position has a positive effect on pulmonary ventilation after exercise and may be the most effective posture to promote recovery of breathing after maximal exertion.

Pulmonary ventilation increases during exercise in direct proportion to the metabolic needs of exercising muscle. At low exercise intensities, this is accomplished by increases in tidal volume (the amount of air moved in and out of the lungs during regular breathing). At higher intensities, the rate of respiration also increases. Maximal rates of pulmonary ventilation depend on body size. Maximal ventilation rates of approximately 100 L/min are common for smaller individuals but may exceed 200 L/min in larger individuals.

At the end of exercise, the muscles' energy demands decrease almost immediately to resting levels. But pulmonary ventilation returns to normal at a slower rate. If the rate of breathing perfectly

matched the metabolic demands of the tissues, respiration would decrease to the resting level within seconds after exercise. But respiratory recovery takes several minutes, which suggests that postexercise breathing is regulated primarily by acid–base balance, the partial pressure of dissolved carbon dioxide (PCO_2), and blood temperature.

Breathing Irregularities During Exercise

Ideally, breathing during exercise is regulated in a way that maximizes aerobic performance. However, respiratory dysfunction during exercise can hinder performance.

Dyspnea

The sensation of **dyspnea** (shortness of breath) during exercise is common among individuals with poor aerobic fitness levels who attempt to exercise at intensities that significantly elevate arterial CO_2 and H^+ concentrations. As discussed in [chapter 7](#), both stimuli send strong signals to the inspiratory center to increase the rate and depth of ventilation. Although exercise-induced dyspnea is sensed as an inability to breathe, the underlying cause is an inability to adjust breathing to blood PCO_2 and H^+ .

Failure to reduce these stimuli during exercise appears to be related to poor conditioning of respiratory muscles. Despite a strong neural drive to ventilate the lungs, the respiratory muscles fatigue easily and are unable to reestablish normal homeostasis.

Exercise-Induced Asthma

In healthy humans, the respiratory system in general and the ability to conduct efficient gas exchange at the lungs in particular do not normally limit exercise performance. However, it is estimated that up to 55% of elite athletes participating in endurance winter sports and swimming experience symptoms of exercise-induced asthma (EIA), exercise-induced bronchospasm (EIB), or both.^{1,6} Exercise-induced asthma is defined as a lower airway obstruction with symptoms that include coughing, wheezing, or dyspnea that is induced by exercise in individuals with underlying asthma. In addition to EIA, EIB is a reduction in lung function measured by the forced expiratory volume in one second (FEV_1) performed after a standardized exercise test.

Many athletes experience these respiratory symptoms, and the onset can occur during childhood or later in life during their sport careers.



Physiologically, there are several different mechanisms by which EIA and EIB may occur in athletes. The classic reasoning has been that hyperventilation during intense exercise leads to increased evaporation of water from the airway surface. This is a result of having to humidify and warm the air coming into the lung coupled with an increased rate of ventilation during intense exercise. The evaporation of water leads to an increase in osmolality, providing a stimulus for water to move from inside cells to the extracellular fluid.

This shrinkage of cells then induces inflammation, in turn causing the airways to constrict.

Other proposed contributors to EIA and EIB in athletes include a disruption to the airway epithelium and microvasculature injury induced by strenuous exercise and airway cooling. Airway cooling causes a reflex increase in parasympathetic nerve activity, causing bronchoconstriction and vasoconstriction of the blood vessels in the bronchioles in order to conserve heat.

Some aspects of EIA and EIB in elite athletes relate to the specific environmental and sport-specific training conditions in which symptoms occur. For example, the cold and dry air that accompanies winter sports,⁶ the ultrafine airborne particles emitted from ice resurfacing machines in indoor ice rinks,¹⁴ the pollen and pollutant exposure in athletes practicing outdoors,² and chemical exposure in chlorine-rich atmospheres for swimmers have all been implicated as causal factors in breathing problems of athletes.

Hyperventilation

The anticipation of or anxiety about exercise, as well as some respiratory disorders, can cause an increase in ventilation in excess of that needed to support exercise. Such overbreathing is termed **hyperventilation**. At rest, hyperventilation can decrease the normal PCO_2 of 40 mmHg in the alveoli and arterial blood to about 15 mmHg. As arterial CO_2 concentrations decrease, blood pH increases. These effects combine to reduce the ventilatory drive. Because the blood leaving the lungs is almost always about 98% saturated with oxygen, an increase in the alveolar PO_2 does not increase the oxygen content of the blood. Consequently, the reduced drive to breathe—along with the improved ability to hold one's breath after hyperventilating—results from carbon dioxide unloading rather than increased blood oxygen. This is sometimes referred to as “blowing off CO_2 .” Even when performed for only a few seconds, such deep, rapid breathing can lead to light-headedness and even loss of consciousness. This phenomenon reveals the sensitivity of the respiratory system's regulation by carbon dioxide and pH.

Valsalva Maneuver

The Valsalva maneuver is a potentially dangerous respiratory procedure that frequently accompanies certain types of exercise, in particular the lifting of heavy objects. This occurs when the individual

- closes the glottis (the opening between the vocal cords),
- increases the intra-abdominal pressure by forcibly contracting the diaphragm and the abdominal muscles, and
- increases the intrathoracic pressure by forcibly contracting the respiratory muscles.

As a result of these actions, air is trapped and pressurized in the lungs. The high intra-abdominal and intrathoracic pressures restrict venous return by collapsing the great veins. This maneuver, if held for an extended period of time, can greatly reduce the volume of blood returning to the heart, decreasing cardiac output and lowering arterial blood pressure. Although the Valsalva maneuver can be helpful in certain circumstances, the maneuver can be dangerous and should be avoided.

Ventilation and Energy Metabolism

During long periods of mild steady-state activity, ventilation matches the rate of energy metabolism, varying in proportion to the volume of oxygen consumed and the volume of carbon dioxide produced ($\dot{V}O_2$ and $\dot{V}CO_2$, respectively) by the body.

Ventilatory Equivalent for Oxygen

The ratio between the volume of air expired or ventilated ($\dot{V}_E$) and the amount of oxygen consumed by the tissues ($\dot{V}O_2$) in a given amount of time is referred to as the **ventilatory equivalent for oxygen** ($\dot{V}$). It is typically measured in liters of air breathed per liter of oxygen consumed per minute.

At rest, the $\dot{V}_E/\dot{V}O_2$ can range from 23 to 28 L of air per liter of oxygen. This value changes very little during mild exercise, such as walking. But when exercise intensity increases to near-maximal levels, the $\dot{V}_E/\dot{V}O_2$ can be greater than 30 L of air per liter of oxygen consumed. In general, however, the $\dot{V}_E/\dot{V}O_2$ remains relatively constant over a wide range of exercise intensities, indicating that the

control of breathing is properly matched to the body's demand for oxygen.

Ventilatory Threshold

As exercise intensity increases, at some point ventilation increases disproportionately to oxygen consumption. The point at which this occurs, typically between ~55% and 70% of $\dot{V}O_{2\max}$, is called the **ventilatory threshold**, illustrated in [figure 8.15](#). At approximately the same intensity as the ventilatory threshold, more lactate starts to appear in the blood. This may result from greater production of lactate or less clearance of lactate or both. This lactic acid combines with sodium bicarbonate (which buffers acid) and forms sodium lactate, water, and carbon dioxide. As we know, the increase in carbon dioxide stimulates chemoreceptors that signal the inspiratory center to increase ventilation. Thus, the ventilatory threshold reflects the respiratory response to increased carbon dioxide levels. Ventilation increases dramatically beyond the ventilatory threshold, as seen in [figure 8.15](#).

The disproportionate increase in ventilation without an equivalent increase in oxygen consumption led to early speculation that the ventilatory threshold might be related to the lactate threshold (that point at which blood lactate production exceeds lactate reuptake and clearance as described in [chapter 5](#)). The ventilatory threshold reflects a disproportionate increase in the volume of carbon dioxide produced per minute ($\dot{V}CO_2$) relative to the oxygen consumed. Recall from [chapter 5](#) that the respiratory exchange ratio (RER) is the ratio of carbon dioxide production to oxygen consumption. Thus, the disproportionate increase in carbon dioxide production also causes RER to increase.

The increased $\dot{V}CO_2$ was thought to result from excess carbon dioxide being released from bicarbonate buffering of lactic acid. Wasserman and McIlroy¹⁸ coined the term **anaerobic threshold** to refer to this phenomenon because they assumed that the sudden increase in CO_2 reflected a shift toward more anaerobic metabolism. They believed that this was a good noninvasive alternative to blood sampling for detecting the onset of anaerobic metabolism. It should

be noted that a number of scientists objected to their use of the term *anaerobic threshold* to refer to this respiratory phenomenon.

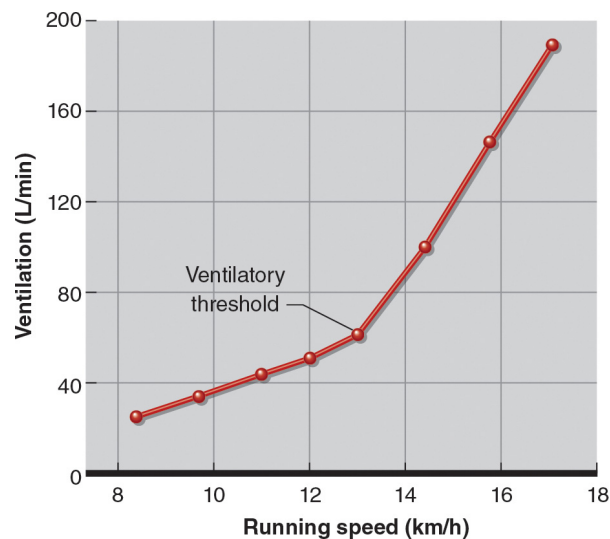


FIGURE 8.15 Changes in pulmonary ventilation ($\dot{V}_E$) during running at increasing velocities, illustrating the concept of ventilatory threshold.

Over the years, the anaerobic threshold concept has been refined considerably to provide a relatively accurate estimate of lactate threshold. One of the more accurate techniques for identifying this threshold involves monitoring both the ventilatory equivalent for oxygen ($\dot{V}_E/\dot{V}O_2$) and the **ventilatory equivalent for carbon dioxide** ($\dot{V}_E/\dot{V}CO_2$), which is the ratio of the volume of air expired ($\dot{V}_E$) to the volume of carbon dioxide produced ($\dot{V}CO_2$). Using this technique, the threshold is defined as that point where there is a systematic increase in $\dot{V}_E/\dot{V}O_2$ without a concomitant increase in $\dot{V}_E/\dot{V}CO_2$. This is illustrated in [figure 8.16](#). Both the $\dot{V}_E/\dot{V}CO_2$ and $\dot{V}_E/\dot{V}O_2$ decline with increasing exercise intensity at the lower intensities. However, the $\dot{V}_E/\dot{V}O_2$ starts to increase at about 75 W while the $\dot{V}_E/\dot{V}CO_2$ continues to decline. This indicates that the increase in ventilation to remove CO_2 is disproportionate to the body's need to provide O_2 . In general, this respiratory threshold technique provides a reasonably close estimate of the lactate threshold, eliminating the need for repeated blood sampling.

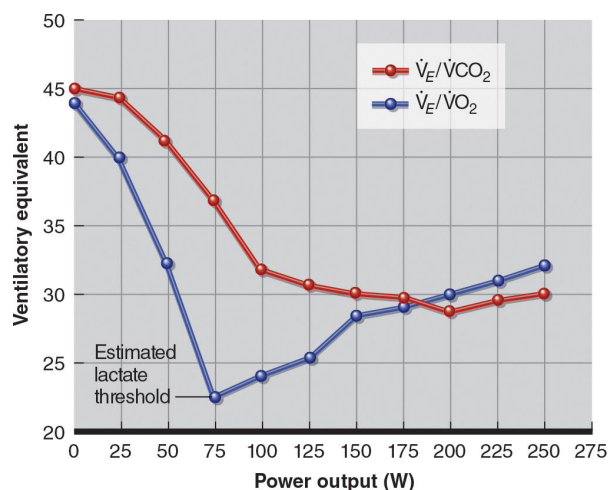


FIGURE 8.16 Changes in the ventilatory equivalent for carbon dioxide ($\dot{V}_E/\dot{V}_{CO_2}$) and the ventilatory equivalent for oxygen ($\dot{V}_E/\dot{V}_{O_2}$) during increasing intensities of exercise on a cycle ergometer. Note that the breakpoint of the estimated lactate threshold at a power output of 75 W is evident only in the $\dot{V}_E/\dot{V}_{O_2}$ ratio.

Respiratory Limitations to Performance

Like all tissue activity, respiration requires energy. Most of this energy is used by the respiratory muscles during pulmonary ventilation. At rest, the respiratory muscles account for only about 2% of the total oxygen uptake. As the rate and depth of ventilation increase, so does the energy cost of respiration. The diaphragm, the intercostal muscles, and the abdominal muscles can account for up to 11% of the total oxygen consumed during heavy exercise and can receive up to 15% of the cardiac output. During recovery from dynamic exercise, sustained elevations in ventilation continue to demand increased energy, accounting for 9% to 12% of the total oxygen consumed postexercise.

In Review

- During exercise, ventilation shows an almost immediate increase due to increased inspiratory center stimulation. This is caused by both central command and neural feedback from muscle activity itself. This phase is followed by a plateau (during light exercise) or a much more gradual increase in respiration (during heavy exercise) that results from chemical changes in the arterial blood resulting from exercise metabolism.

- Altered breathing patterns and sensations associated with exercise include dyspnea, exercise-induced asthma or bronchospasm, hyperventilation, and performance of the Valsalva maneuver.
 - During mild, steady-state exercise, ventilation increases to match the rate of energy metabolism; that is, ventilation parallels oxygen uptake. The ratio of air ventilated to oxygen consumed is the ventilatory equivalent for oxygen ($\dot{V}_E/\dot{V}O_2$).
 - At low exercise intensities, increased ventilation is accomplished by increases in tidal volume (the amount of air moved in and out of the lungs during regular breathing). At higher intensities, the rate of respiration also increases.
 - Maximal rates of pulmonary ventilation depend on body size. Maximal ventilation rates of approximately 100 L/min are common for smaller individuals but may exceed 200 L/min in larger individuals.
 - The ventilatory threshold is the point at which ventilation begins to increase disproportionately to the increase in oxygen consumption. This increase in $\dot{V}_E$ reflects the need to remove excess carbon dioxide.
 - We can estimate lactate threshold with reasonable accuracy by identifying that point at which $\dot{V}_E/\dot{V}O_2$ starts to increase while $\dot{V}_E/\dot{V}CO_2$ continues to decline.
-

Although the muscles of respiration are heavily taxed during exercise, ventilation is sufficient to prevent an increase in alveolar PCO_2 or a decline in alveolar PO_2 during activities lasting only a few minutes. Even during maximal effort, ventilation usually is not pushed to its maximal capacity to voluntarily move air in and out of the lungs. This capacity is called the **maximal voluntary ventilation** and is significantly greater than ventilation at maximal exercise. However, considerable evidence suggests that pulmonary ventilation might be a limiting factor during exercise of very high intensity (95%-100% $\dot{V}O_{2max}$) in highly trained subjects.

Can heavy breathing for several hours (such as during marathon running) cause glycogen depletion and fatigue of the respiratory muscles? Animal studies have shown a substantial sparing of their respiratory muscle glycogen compared with muscle glycogen in exercising muscles. Although similar data are not available for humans, our respiratory muscles are better designed for long-term activity than are the muscles in our extremities. The diaphragm, for example, has two to three times more oxidative capacity (oxidative enzymes and mitochondria) and capillary density than other skeletal

muscle. Consequently, the diaphragm can obtain more energy from oxidative sources than can skeletal muscles.

Similarly, airway resistance and gas diffusion in the lungs do not limit exercise in a normal, healthy individual. The volume of air inspired can increase 20- to 40-fold with exercise—from ~5 L/min at rest up to 100 to 200 L/min with maximal exertion. Airway resistance, however, is maintained at near-resting levels by airway dilation (through an increase in the laryngeal aperture and bronchodilation). During submaximal and maximal efforts in untrained and moderately trained individuals, blood leaving the lungs remains nearly saturated with oxygen (~98%). However, with maximal exercise in some highly trained elite endurance athletes, there is too large a demand on lung gas exchange, resulting in a decline in arterial PO_2 and arterial oxygen saturation (i.e., **exercise-induced arterial hypoxemia [EIAH]**). Approximately 40% to 50% of elite endurance athletes experience a significant reduction in arterial oxygenation during exercise approaching exhaustion.¹⁰ Arterial hypoxemia at maximal exercise is likely the result of a mismatch between ventilation and perfusion of the lung. Since cardiac output is extremely high in elite athletes, blood is flowing through the lungs at a high rate and thus there may not be sufficient time for that blood to become saturated with oxygen. Thus, in healthy individuals, the respiratory system is well designed to accommodate the demands of heavy breathing during short- and long-term physical effort. However, some highly trained individuals who consume unusually large amounts of oxygen during exhaustive exercise can face respiratory limitations.

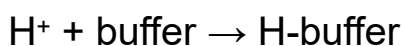
The respiratory system also can limit performance in patient populations with restricted or obstructed airways. For example, asthma causes constriction of the bronchial tubes and swelling of the mucous membranes. These effects cause considerable resistance to ventilation, resulting in a shortness of breath. Exercise is known to bring about symptoms of asthma or to worsen those symptoms in select individuals. The mechanism or mechanisms through which exercise induces airway obstruction in individuals with so-called exercise-induced asthma remain unknown, despite extensive study.

- Respiratory muscles can account for up to 10% of the body's total oxygen consumption and 15% of the cardiac output during heavy exercise.
 - Pulmonary ventilation is usually not a limiting factor for performance even during maximal effort, although it can limit performance in some elite endurance athletes.
 - The respiratory muscles are well designed to avoid fatigue during long-term activity.
 - Airway resistance and gas diffusion usually do not limit performance in normal, healthy individuals exercising at sea level.
 - The respiratory system can, and often does, limit performance in people with various types of restrictive or obstructive respiratory disorders.
-

Respiratory Regulation of Acid–Base Balance

As noted earlier, high-intensity exercise results in the production and accumulation of lactate and H^+ . Although regulation of acid–base balance involves more than control of respiration, it is discussed here because the respiratory system plays such a crucial role in rapid adjustment of the body's acid–base status during and immediately after exercise.

Acids, such as lactic acid and carbonic acid, release hydrogen ions (H^+). As noted in the preceding chapters, the metabolism of carbohydrate, fat, or protein produces inorganic acids that dissociate, increasing the H^+ concentration in body fluids, thus lowering the pH. To minimize the effects of free H^+ , the blood and muscles contain base substances that combine with, and thus buffer or neutralize, the H^+ :



Under resting conditions, body fluids have more bases (such as bicarbonate, phosphate, and proteins) than acids, resulting in a slightly alkaline tissue pH that ranges from 7.1 in muscle to 7.4 in arterial blood. The tolerable limits for arterial blood pH extend from 6.9 to 7.5, although the extremes of this range can be tolerated only for a few minutes (see [figure 8.17](#)). An H^+ concentration above normal (low pH) is referred to as acidosis, whereas a decrease in H^+ below the normal concentration (high pH) is termed alkalosis.

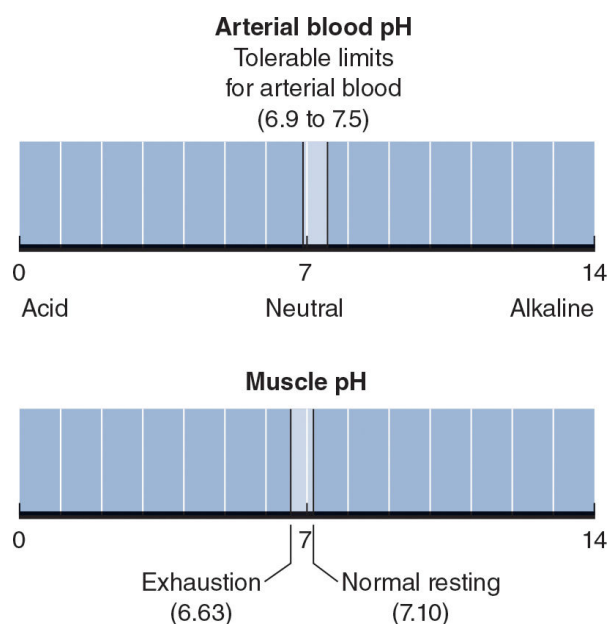


FIGURE 8.17 Tolerable limits for arterial blood pH and muscle pH at rest and at exhaustion. Note the small range of physiological tolerance for both muscle and blood pH.

The pH of intra- and extracellular body fluids is kept within a relatively narrow range by

- chemical buffers in the blood,
- pulmonary ventilation, and
- kidney function.

The three major chemical buffers in the body are bicarbonate (HCO_3^-), inorganic phosphates (P_i), and proteins. In addition to these, hemoglobin in the red blood cells is also a major buffer. [Table 8.2](#) illustrates the relative contributions of these buffers in handling acids in the blood. Recall that bicarbonate combines with H^+ to form carbonic acid, thereby eliminating the acidifying influence of free H^+ . The carbonic acid in turn forms carbon dioxide and water in the lungs. The CO_2 is then exhaled and only water remains.

TABLE 8.2 Buffering Capacity of Blood Components

Buffer	Slykesa	%
Bicarbonate	18.0	64
Hemoglobin	8.0	29
Proteins	1.7	6
Phosphates	0.3	1
Total	28.0	100

^aMilliequivalents of hydrogen ions taken up by each liter of blood from pH 7.4 to 7.0.

The amount of bicarbonate that combines with H^+ equals the amount of acid buffered. When lactic acid decreases the blood's pH from 7.4 to 7.0, more than 60% of the bicarbonate initially present in the blood has been used. Even under resting conditions, the acid produced by the end products of metabolism would use up a major portion of the bicarbonate from the blood if there were no other way of removing H^+ from the body. Blood and chemical buffers are required only to transport metabolic acids from their sites of production (the muscles) to the lungs or kidneys, where they can be removed. Once H^+ is transported and removed, the buffer molecules can be reused.

In the muscle fibers and the kidney tubules, H^+ is primarily buffered by phosphates, such as phosphoric acid and sodium phosphate. Less is known about the capacity of the buffers intracellularly, although cells contain more protein and phosphates and less bicarbonate than do the extracellular fluids.

As noted earlier, any increase in free H^+ in the blood stimulates the respiratory center to increase ventilation. This facilitates the binding of H^+ to bicarbonate and the removal of carbon dioxide. The end result is a decrease in free H^+ and an increase in blood pH. Thus, both the chemical buffers and the respiratory system provide short-term means of neutralizing the acute effects of exercise acidosis. To maintain a constant buffer reserve, the accumulated H^+ is removed from the body via excretion by the kidneys and eliminated in urine. The kidneys filter H^+ from the blood along with other waste products. This provides a way to eliminate H^+ from the body while maintaining the concentration of extracellular bicarbonate.



During sprint exercise, muscle glycolysis generates a large amount of lactate and H^+ , which lowers the muscle pH from a resting level of 7.1 to less than 6.7. As shown in [table 8.3](#), an all-out 400 m sprint decreases leg muscle pH to 6.63 and increases muscle lactate from a resting value of 1.2 mmol/kg to almost 20 mmol/kg of muscle. Such disturbances in acid–base balance can impair muscle contractility and its capacity to generate adenosine triphosphate (ATP). Lactate and H^+ accumulate in the muscle, in part because they do not freely diffuse across the skeletal muscle fiber membranes. Despite the great production of lactate and H^+ during the ~60 s required to run 400 m, these by-products diffuse throughout the body fluids and reach equilibrium after only about 5 to 10 min of recovery. Five minutes after the exercise, the runners described in [table 8.3](#) had blood pH values of 7.10 and blood lactate concentrations of 12.3 mmol/L, compared with a resting pH of 7.40 and a resting lactate level of 1.5 mmol/L.

TABLE 8.3 Blood and Muscle pH and Lactate Concentration 5 min After a 400 m Run

Runner	Time (s)	Muscle		Blood	
		pH	Lactate (mmol/kg)	pH	Lactate (mmol/L)
1	61.0	6.68	19.7	7.12	12.6
2	57.1	6.59	20.5	7.14	13.4
3	65.0	6.59	20.2	7.02	13.1
4	58.5	6.68	18.2	7.10	10.1
Average	60.4	6.64	19.7	7.10	12.3

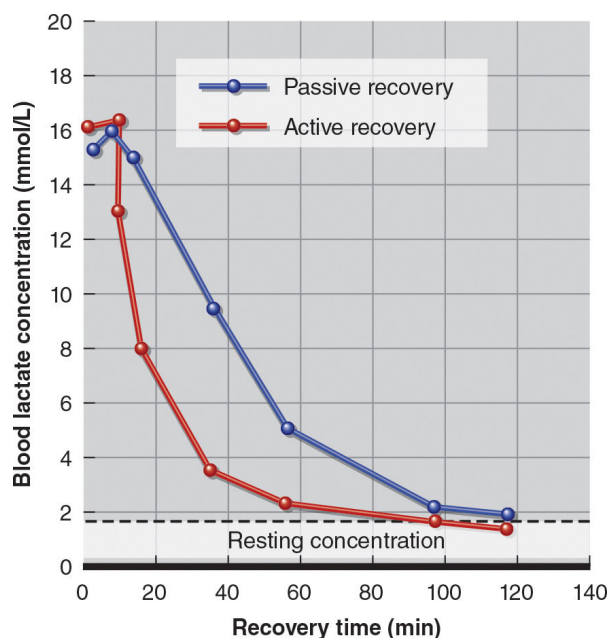


FIGURE 8.18 Effects of active and passive recovery on blood lactate concentrations after a series of exhaustive sprint bouts. Note that the blood lactate removal rate is faster when the subjects perform exercise during recovery than when they rest during recovery.

Reestablishing normal resting concentrations of blood and muscle lactate after such an exhaustive exercise bout is a relatively slow process, often requiring 1 to 2 h. As shown in [figure 8.18](#), recovery of blood lactate to the resting level is facilitated by continued lower-intensity exercise, called active recovery.³ After a series of exhaustive sprint bouts, the participants in this study either sat quietly (passive recovery) or exercised at an intensity of 50% $\dot{V}O_{2\max}$. Blood lactate is removed more quickly during active recovery because the activity maintains elevated blood flow through the active muscles, which in turn enhances both lactate diffusion out of the muscles and lactate oxidation.

Although blood lactate remains elevated for 1 to 2 h after highly anaerobic exercise, blood and muscle H^+ concentrations return to normal within 40 min of recovery. Chemical buffering, principally by bicarbonate, and respiratory removal of excess carbon dioxide are responsible for this relatively rapid return to normal acid–base homeostasis.

In Review

- Excess H^+ (decreased pH) impairs muscle contractility and ATP generation.

- The respiratory and renal systems play integral roles in maintaining acid–base balance. The renal system is involved in more long-term maintenance of acid–base balance through the secretion of H^+ .
 - Whenever H^+ concentration starts to increase, the inspiratory center responds by increasing the rate and depth of respiration. Removing carbon dioxide is an essential means of reducing H^+ concentrations.
 - Carbon dioxide is transported in the blood primarily bound to bicarbonate. Once it reaches the lungs, carbon dioxide is formed again and exhaled.
 - Whenever H^+ concentration begins to increase, whether from carbon dioxide or lactate accumulation, bicarbonate ion can buffer the H^+ to prevent acidosis.
-

IN CLOSING

In this chapter, we discussed the responses of the cardiovascular and respiratory systems to exercise. We also considered the limitations that these systems can impose on abilities to perform sustained aerobic exercise. The next chapter presents basic principles of exercise training, allowing us to better understand in the subsequent chapters how the body adapts to resistance training as well as aerobic and anaerobic training.

KEY TERMS

afterload
anaerobic threshold
cardiovascular drift
central command
dyspnea
exercise-induced arterial hypoxemia (EIAH)
Frank-Starling mechanism
hydrostatic pressure
hyperventilation
maximal voluntary ventilation
maximum heart rate ($HR_{\max}$)
oncotic pressure
preload
rate–pressure product (RPP)
resting heart rate (RHR)
steady-state heart rate
total peripheral resistance (TPR)
Valsalva maneuver
ventilatory equivalent for carbon dioxide ($\dot{V}_E/\dot{V}CO_2$)
ventilatory equivalent for oxygen ($\dot{V}_E/\dot{V}O_2$)
ventilatory threshold

STUDY QUESTIONS

1. Describe how heart rate, stroke volume, and cardiac output respond to increasing rates of work. Illustrate how these three variables are interrelated.
2. How do we determine $HR_{\max}$? What are alternative methods using indirect estimates? What are the major limitations of these indirect estimates?
3. What information can be learned from measuring heart rate variability?

4. Describe two important mechanisms for returning blood back to the heart during exercise in an upright position.
5. Explain why the ability to increase stroke volume is important in determining maximal oxygen consumption.
6. What is the Fick principle, and how does this apply to our understanding of the relationship between metabolism and cardiovascular function?
7. Define the Frank-Starling mechanism. How does this work during exercise?
8. How does blood pressure respond to exercise?
9. What are the major cardiovascular adjustments that the body makes when someone is overheated during exercise?
10. What is cardiovascular drift? What two theories have been proposed to explain this phenomenon?
11. What changes occur in the plasma volume and red blood cells with increasing levels of exercise? With prolonged exercise in the heat?
12. How does pulmonary ventilation respond to increasing intensities of exercise?
13. Define the terms *dyspnea*, *hyperventilation*, *Valsalva maneuver*, and *ventilatory threshold*.
14. What causes exercise-induced asthma in some athletes? What athletes are most prone to being affected?
15. What role does the respiratory system play in acid–base balance?
16. What is the normal resting pH for arterial blood? For muscle? How are these values changed as a result of exhaustive sprint exercise?
17. What are the primary buffers in the blood? In muscles?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter QUIZ tests your understanding of the material covered in the chapter.